

Noninvasive Treatments for Acute, Subacute, and Chronic Low Back Pain: A Clinical Practice Guideline From the American College of Physicians

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Description: The American College of Physicians (ACP) developed this guideline to present the evidence and provide clinical recommendations on noninvasive treatment of low back pain.

Methods: Using the ACP grading system, the committee based these recommendations on a systematic review of randomized, controlled trials and systematic reviews published through April 2015 on noninvasive pharmacologic and nonpharmacologic treatments for low back pain. Updated searches were performed through November 2016. Clinical outcomes evaluated included reduction or elimination of low back pain, improvement in back-specific and overall function, improvement in health-related quality of life, reduction in work disability and return to work, global improvement, number of back pain episodes or time between episodes, patient satisfaction, and adverse effects.

Target Audience and Patient Population: The target audience for this guideline includes all clinicians, and the target patient population includes adults with acute, subacute, or chronic low back pain.

Recommendation 1: *Given that most patients with acute or subacute low back pain improve over time regardless of treatment, clinicians and patients should select nonpharmacologic treatment with superficial heat (moderate-quality evidence), massage, acupuncture, or spinal manipulation (low-quality evidence). If pharmacologic treatment is desired, clinicians and patients should select nonsteroidal anti-inflammatory drugs or skeletal*

muscle relaxants (moderate-quality evidence). (Grade: strong recommendation)

Recommendation 2: *For patients with chronic low back pain, clinicians and patients should initially select nonpharmacologic treatment with exercise, multidisciplinary rehabilitation, acupuncture, mindfulness-based stress reduction (moderate-quality evidence), tai chi, yoga, motor control exercise, progressive relaxation, electromyography biofeedback, low-level laser therapy, operant therapy, cognitive behavioral therapy, or spinal manipulation (low-quality evidence). (Grade: strong recommendation)*

Recommendation 3: *In patients with chronic low back pain who have had an inadequate response to nonpharmacologic therapy, clinicians and patients should consider pharmacologic treatment with nonsteroidal anti-inflammatory drugs as first-line therapy, or tramadol or duloxetine as second-line therapy. Clinicians should only consider opioids as an option in patients who have failed the aforementioned treatments and only if the potential benefits outweigh the risks for individual patients and after a discussion of known risks and realistic benefits with patients. (Grade: weak recommendation, moderate-quality evidence)*

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Low back pain is one of the most common reasons for physician visits in the United States. Most Americans have experienced low back pain, and approximately one quarter of U.S. adults reported having low back pain lasting at least 1 day in the past 3 months (1).

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Low back pain is associated with high costs, including those related to health care and indirect costs from missed work or reduced productivity (2). The total costs attributable to low back pain in the United States were estimated at \$100 billion in 2006, two thirds of which were indirect costs of lost wages and productivity (3).

Low back pain is frequently classified and treated on the basis of symptom duration, potential cause, presence or absence of radicular symptoms, and corresponding anatomical or radiographic abnormalities. Acute back pain is defined as lasting less than 4 weeks, subacute back pain lasts 4 to 12 weeks, and chronic back pain lasts more than 12 weeks. Radicular low back pain results in lower extremity pain, paresthesia, and/or

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weakness and is a result of nerve root impingement. Most patients with acute back pain have self-limited episodes that resolve on their own; many do not seek medical care (4). For patients who do seek medical care, pain, disability, and return to work typically improve rapidly in the first month (5). However, up to one third of patients report persistent back pain of at least moderate intensity 1 year after an acute episode, and 1 in 5 report substantial limitations in activity (6). Many noninvasive treatment options are available for radicular and nonradicular low back pain, including pharmacologic and nonpharmacologic interventions.

GUIDELINE FOCUS AND TARGET POPULATION

The purpose of this American College of Physicians (ACP) guideline is to provide treatment guidance based on the efficacy, comparative effectiveness, and safety of noninvasive pharmacologic and nonpharmacologic treatments for acute (<4 weeks), subacute (4 to 12 weeks), and chronic (>12 weeks) low back pain in primary care. This guideline does not address topical pharmacologic therapies or epidural injection therapies. It serves as a partial update of the 2007 ACP guideline (it excludes evidence on diagnosis). These recommendations are based on 2 background evidence reviews (7, 8) and a systematic review sponsored by the Agency for Healthcare Research and Quality (AHRQ) (9). The target audience for this guideline includes all clinicians, and the target patient population includes adults with acute, subacute, or chronic low back pain.

METHODS

Systematic Review of the Evidence

The evidence review was conducted by the AHRQ's Pacific Northwest Evidence-based Practice Center. Additional methodological details can be found in the **Appendix** (available at Annals.org) as well as in the accompanying articles (7, 8) and full report (9). Reviewers searched several databases for studies published in English from January 2008 through April 2015 and updated the search through November 2016. Studies published before 2007 were identified using the 2007 ACP/American Pain Society (APS) systematic reviews (10, 11). Reviewers combined data when possible using meta-analysis and assessed risk of bias and study quality according to established methods. The study population included adults (aged ≥ 18 years) with acute, subacute, or chronic nonradicular low back pain, radicular low back pain, or symptomatic spinal stenosis.

The review evaluated pharmacologic (acetaminophen, nonsteroidal anti-inflammatory drugs [NSAIDs], opioids, skeletal muscle relaxants [SMRs], benzodiazepines, antidepressants, antiseizure medications, and systemic corticosteroids) and nonpharmacologic (psychological therapies, multidisciplinary rehabilitation, spinal manipulation, acupuncture, massage, exercise and related therapies, and various physical modalities) treatments for low back pain. Evaluated outcomes in-

Table. The American College of Physicians Guideline Grading System*

Quality of Evidence	Strength of Recommendation	
	Benefits Clearly Outweigh Risks and Burden or Risks and Burden Clearly Outweigh Benefits	Benefits Finely Balanced With Risks and Burden
High	Strong	Weak
Moderate	Strong	Weak
Low	Strong	Weak
Insufficient evidence to determine net benefits or risks		

* Adopted from the classification developed by the GRADE (Grading of Recommendations Assessment, Development and Evaluation) workgroup.

cluded reduction or elimination of low back pain, improvement in back-specific and overall function, improvement in health-related quality of life, reduction in work disability, return to work, global improvement, number of back pain episodes or time between episodes, patient satisfaction, and adverse effects.

The magnitude of effect (small, moderate, or large) was determined as previously described (10, 11). A small effect on pain was defined as a mean between-group difference after treatment of 5 to 10 points on a visual analogue scale of 0 to 100 or equivalent, a mean between-group difference of 0.5 to 1.0 point on a numerical rating scale of 0 to 10, or a standardized mean difference of 0.2 to 0.5. A moderate effect was defined as a mean between-group difference of greater than 10 to no more than 20 points on a visual analogue scale of 0 to 100 or equivalent, a mean between-group difference of greater than 1.0 to no more than 2.0 points on a numerical rating scale of 0 to 10 or equivalent, or a standardized mean difference greater than 0.5 but no more than 0.8. For function, a small effect was defined as a mean between-group difference of 5 to 10 points on the Oswestry Disability Index (ODI), a mean between-group difference of 1 to 2 points on the Roland Morris Disability Questionnaire (RDQ), or a standardized mean difference of 0.2 to 0.5. A moderate effect on function was defined as a mean between-group difference of greater than 10 to no more than 20 points on the ODI, a mean between-group difference of greater than 2 to no more than 5 points on the RDQ, or a standardized mean difference greater than 0.5 but no more than 0.8. No large effects were found with any intervention.

Grading the Evidence and Developing Recommendations

This guideline was developed by ACP's Clinical Guidelines Committee (CGC) according to ACP's guideline development process, details of which can be found in the methods paper (12). The CGC used the evidence tables in the accompanying evidence reviews (7, 8) and full report (9) when reporting the evidence

and graded the recommendations using the ACP's guideline grading system (Table).

Peer Review

The AHRQ systematic review was sent to invited peer reviewers and posted on the AHRQ Web site for public comments. The accompanying evidence reviews (7, 8) also underwent a peer review process through the journal. The guideline underwent a peer review process through the journal and was posted online for comments from ACP Regents and ACP Governors, who represent ACP members at the regional level.

BENEFITS AND COMPARATIVE BENEFITS OF PHARMACOLOGIC THERAPIES

Acute or Subacute Low Back Pain

Appendix Table 1 (available at [Annals.org](#)) summarizes the findings for all therapies for acute or subacute low back pain.

Acetaminophen

Low-quality evidence showed no difference between acetaminophen and placebo for pain intensity or function through 4 weeks or between acetaminophen and NSAIDs for pain intensity or likelihood of experiencing global improvement at 3 weeks or earlier (13, 14).

NSAIDs

Moderate-quality evidence showed that NSAIDs were associated with a small improvement in pain intensity compared with placebo (14, 15), although several randomized, controlled trials (RCTs) showed no difference in likelihood of achieving pain relief with NSAIDs compared with placebo (16–18). Low-quality evidence showed a small increase in function with NSAIDs compared with placebo (19). Moderate-quality evidence showed that most head-to-head trials of one NSAID versus another showed no differences in pain relief in patients with acute low back pain (14). Low-quality evidence showed no differences in pain between cyclooxygenase (COX)-2-selective NSAIDs versus traditional NSAIDs (14).

SMRs

Moderate-quality evidence showed that SMRs improved short-term pain relief compared with placebo after 2 to 4 and 5 to 7 days (20, 21). Low-quality evidence showed no differences between different SMRs for any outcomes in patients with acute pain (20). Low-quality evidence showed inconsistent findings for the effect on pain intensity with a combination of SMRs plus NSAIDs compared with NSAIDs alone (20, 22, 23).

Systemic Corticosteroids

Low-quality evidence showed no difference in pain or function between a single intramuscular injection of

methylprednisolone or a 5-day course of prednisolone compared with placebo in patients with acute low back pain (24, 25).

Other Therapies

Evidence was insufficient to determine effectiveness of antidepressants, benzodiazepines (26, 27), antiseizure medications, or opioids versus placebo in patients with acute or subacute low back pain.

Chronic Low Back Pain

Appendix Table 2 (available at [Annals.org](#)) summarizes the findings for all therapies for chronic low back pain.

NSAIDs

Moderate-quality evidence showed that NSAIDs were associated with small to moderate pain improvement compared with placebo (14, 28, 29). Low-quality evidence showed that NSAIDs were associated with no to small improvement in function (28–31). Moderate-quality evidence showed that most head-to-head trials of one NSAID versus another showed no differences in pain relief in patients with chronic low back pain (14). There were no data on COX-2-selective NSAIDs.

Opioids

Moderate-quality evidence showed that strong opioids (tapentadol, morphine, hydromorphone, and oxycodone) were associated with a small short-term improvement in pain scores (about 1 point on a pain scale of 0 to 10) and function compared with placebo (32–36). Low-quality evidence showed that buprenorphine patches improved short-term pain more than placebo in patients with chronic low back pain; however, the improvement corresponded to less than 1 point on a pain scale of 0 to 10 (37–40). Moderate-quality evidence showed no differences among different long-acting opioids for pain or function (33, 41–44), and low-quality evidence showed no clear differences in pain relief between long- and short-acting opioids (45–50). Moderate-quality evidence showed that tramadol achieved moderate short-term pain relief and a small improvement in function compared with placebo (32, 51, 52).

SMRs

Evidence comparing SMRs versus placebo was insufficient (53–55). Low-quality evidence showed no differences in any outcome between different SMRs for treatment of chronic low back pain (20).

Benzodiazepines

Low-quality evidence showed that tetrazepam improved pain relief at 5 to 7 days and resulted in overall

improvement at 10 to 14 days compared with placebo (20).

Antidepressants

Moderate-quality evidence showed no difference in pain between tricyclic antidepressants (TCAs) or selective serotonin reuptake inhibitors (SSRIs) versus placebo, and low-quality evidence showed no differences in function for antidepressants (56). Moderate-quality evidence showed that duloxetine was associated with a small improvement in pain intensity and function compared with placebo (57–59).

Other Therapies

Evidence was insufficient to determine the effect of acetaminophen, systemic corticosteroids, or antiseizure medications on chronic low back pain.

Radicular Low Back Pain

Appendix Table 3 (available at Annals.org) summarizes the findings for all treatments for radicular low back pain.

Benzodiazepines

Low-quality evidence showed no difference between diazepam and placebo for function at 1 week through 1 year and analgesic use, return to work, or likelihood of surgery through 1 year of follow-up in patients with acute or subacute radicular pain (60). Diazepam resulted in a lower likelihood of pain improvement at 1 week compared with placebo.

Systemic Corticosteroids

Moderate-quality evidence showed no differences in pain between systemic corticosteroids and placebo and no to small effect on function in patients with radicular low back pain (61–66).

Other Therapies

No RCTs evaluated acetaminophen, SMRs, antidepressants, or opioids for radicular low back pain. Results for NSAIDs were inconsistent for pain, and evidence was therefore insufficient (22). There was insufficient evidence to determine the effect of antiseizure medications on radicular low back pain (67–71).

HARMS OF PHARMACOLOGIC THERAPIES

Harms were derived from the identified systematic reviews. Adverse effects generally associated with the drugs can be found in Appendix Table 4 (available at Annals.org).

Moderate-quality evidence showed no difference among scheduled acetaminophen, acetaminophen taken as needed, or placebo for serious adverse events (13). Moderate-quality evidence showed that more adverse effects occurred with NSAIDs than placebo, COX-2-selective NSAIDs were associated with a decreased risk for adverse effects compared with traditional

NSAIDs, and acetaminophen was associated with a lower risk for adverse effects than NSAIDs (14). Moderate-quality evidence showed that short-term use of opioids increased nausea, dizziness, constipation, vomiting, somnolence, and dry mouth compared with placebo, and SMRs increased risk for any adverse event and central nervous system adverse events (mostly sedation) compared with placebo (20). Moderate-quality evidence showed that antidepressants increased risk for any adverse event compared with placebo, although rates of specific adverse events did not differ (72). The risk for serious adverse events did not differ between duloxetine and placebo, although duloxetine was associated with increased risk for withdrawal due to adverse events (57–59). Low-quality evidence showed no clear differences in adverse effects for gabapentin versus placebo (67, 68). Low-quality evidence showed that benzodiazepines caused more frequent somnolence, fatigue, and lightheadedness than placebo (20). Harms were not well-reported, and no RCTs assessed long-term use of benzodiazepines or risks for addiction, abuse, or overdose. Adverse events for systemic corticosteroids were not well-reported in RCTs, but the largest trial found that oral prednisone was associated with increased risk for any adverse event, insomnia, nervousness, and increased appetite (66). However, low-quality evidence showed no cases of hyperglycemia that required medical attention (24, 61, 64).

COMPARATIVE BENEFITS OF NONPHARMACOLOGIC THERAPIES

Acute or Subacute Low Back Pain

Exercise

Low-quality evidence showed no difference between exercise therapy and usual care for pain or function in patients with acute or subacute pain (11); additional trials reported inconsistent results (73–75). Moderate-quality evidence showed no clear differences between different exercise regimens in more than 20 head-to-head RCTs in patients with acute low back pain.

Acupuncture

Low-quality evidence showed that acupuncture resulted in a small decrease in pain intensity compared with sham acupuncture with nonpenetrating needles, but there were no clear effects on function (76–78). Low-quality evidence showed that acupuncture slightly increased the likelihood of overall improvement compared with NSAIDs (76, 79–83).

Massage

Low-quality evidence showed that massage moderately improved short-term (1 week) pain and function compared with sham therapy for subacute low back pain (84), although 1 trial (85) showed no difference in pain or function at 5 weeks. Moderate-quality evidence showed that massage improved short-term pain relief

and function compared with other interventions (manipulation, exercise therapy, relaxation therapy, acupuncture, or physiotherapy) for patients with subacute to chronic low back pain, but effects were small (84, 86). Low-quality evidence showed that a combination of massage plus another intervention (exercise, exercise and education, or usual care) was superior to the other intervention alone for short-term pain in patients with subacute to chronic low back pain (84).

Spinal Manipulation

Low-quality evidence showed that spinal manipulation was associated with a small effect on function compared with sham manipulation; evidence was insufficient to determine the effect on pain (87, 88). Low-quality evidence showed no difference in pain relief at 1 week between spinal manipulation and inert treatment (educational booklet, detuned ultrasound, detuned or actual short-wave diathermy, antiedema gel, or bed rest), although 1 trial showed better longer-term pain relief (3 months) with spinal manipulation (89). Function did not differ between spinal manipulation and inert treatment at 1 week or 3 months (89). Moderate-quality evidence showed no difference between spinal manipulation and other active interventions for pain relief at 1 week through 1 year or function (analyses included exercise, physical therapy, or back school as the comparator) (89, 90). Low-quality evidence showed that a combination of spinal manipulation plus exercise or advice slightly improved function at 1 week compared with exercise or advice alone, but these differences were not present at 1 or 3 months (89).

Superficial Heat

Moderate-quality evidence showed that a heat wrap moderately improved pain relief (at 5 days) and disability (at 4 days) compared with placebo (91). Low-quality evidence showed that a combination of heat plus exercise provided greater pain relief and improved RDQ scores at 7 days compared with exercise alone in patients with acute pain (92). Low-quality evidence showed that a heat wrap provided more effective pain relief and improved RDQ scores compared with acetaminophen or ibuprofen after 1 to 2 days (93). Low-quality evidence showed no clear differences between a heat wrap and exercise in pain relief or function (92).

Low-Level Laser Therapy

Low-quality evidence showed that a combination of low-level laser therapy (LLLT) and NSAIDs largely decreased pain intensity and resulted in a moderate improvement in function (as measured by the ODI) compared with sham laser therapy plus NSAIDs in patients with acute or subacute pain (94).

Lumbar Supports

Low-quality evidence showed no difference in pain or function between lumbar supports added to an educational program compared with an educational program alone or other active interventions in patients with acute or subacute low back pain (95).

Other Therapies

Evidence was insufficient to determine the effectiveness of transcutaneous electrical nerve stimulation (TENS), electrical muscle stimulation, inferential therapy, short-wave diathermy, traction, superficial cold, motor control exercise (MCE), Pilates, tai chi, yoga, psychological therapies, multidisciplinary rehabilitation, ultrasound, and taping.

Chronic Low Back Pain

Exercise

Moderate-quality evidence showed that exercise resulted in a small improvement in pain relief and function compared with no exercise (11, 96). Moderate-quality evidence showed that compared with usual care, exercise resulted in small improvements in pain intensity and function at the end of treatment, although effects were smaller at long-term follow-up (96). Moderate-quality evidence showed no clear differences between different exercise regimens in more than 20 head-to-head RCTs in patients with chronic low back pain.

MCE

Motor control exercise focuses on restoring coordination, control, and strength of the muscles that control and support the spine. Low-quality evidence showed that MCE moderately decreased pain scores and slightly improved function in short- to long-term follow-up compared with a minimal intervention (97). Low-quality evidence showed that MCE resulted in small improvements in pain intensity at short-term (≥ 6 weeks to < 4 months) and intermediate-term (≥ 4 to < 8 months) follow-up compared with general exercise, although improvements were small and no longer significant at long-term follow-up (97). Motor control exercise also resulted in small improvements in function in the short and long term (97). Low-quality evidence showed that MCE resulted in a moderate improvement in pain intensity and function compared with multimodal physical therapy at intermediate follow-up (97). Low-quality evidence showed no clear differences in pain with a combination of MCE plus exercise versus exercise alone (98, 99).

Pilates

Low-quality evidence showed that Pilates resulted in small or no clear effects on pain and no clear effects on function compared with usual care plus physical activity (100–107). Low-quality evidence showed no clear differences between Pilates and other types of exercise for pain or function (108–110).

Tai Chi

Low-quality evidence showed that tai chi resulted in moderate pain improvement compared with wait-list controls or no tai chi (111, 112), and 1 study showed a small increase in function (111). Moderate-quality evidence showed that tai chi moderately decreased pain intensity at 3 and 6 months compared with backward walking or jogging but not versus swimming (112).

Yoga

Low-quality evidence showed that Iyengar yoga resulted in moderately lower pain scores and improved function compared with usual care at 24 weeks (113). Low-quality evidence showed that yoga resulted in a small decrease in pain intensity compared with exercise (114–118). Low-quality evidence showed that, compared with education, yoga resulted in a small decrease in short-term (≤ 12 weeks) but not long-term (about 1 year) pain intensity and a small increase in short- and long-term function (119).

Psychological Therapies

Low-quality evidence showed that progressive relaxation therapy moderately improved pain intensity and functional status compared with wait-list controls (120). Low-quality evidence showed that electromyography biofeedback training moderately decreased pain intensity (reduction of 5 to 13 points on a 100-point pain scale) compared with wait-list controls, but there was no effect on function (120). Low-quality evidence showed that operant therapy (behavioral therapy involving reinforcement) slightly improved pain intensity compared with wait-list control, although there was no difference for function (120). Low-quality evidence showed that cognitive behavioral therapy (CBT) and other combined psychological therapies (involving education, problem-solving training, coping techniques, imagery, relaxation, goal setting, cognitive pain control, and exercises) were associated with moderately improved pain intensity compared with wait-list controls, but there was no difference in function (120). Moderate-quality evidence showed that mindfulness-based stress reduction is an effective treatment for chronic low back pain. One study showed a small improvement in pain at 26 and 52 weeks and in function at 26 weeks compared with usual care (121). The same study showed no difference between mindfulness-based stress reduction and CBT for improvements in pain or function. Two other studies showed improvement in pain and function compared with education (122, 123). Low-quality evidence showed no difference between psychological therapies and exercise or physical therapy for pain intensity (120). Low-quality evidence showed no differences in pain or function between a combination of psychological therapy plus exercise or physiotherapy compared with exercise or physiotherapy alone (120). Moderate-quality evidence showed no differences between different psychological therapies for pain or function outcomes (120).

Multidisciplinary Rehabilitation

Moderate-quality evidence showed that multidisciplinary rehabilitation moderately reduced short-term (< 3 months) and slightly reduced long-term pain intensity and disability compared with usual care, although there was no difference in return to work (124). Low-quality evidence showed that multidisciplinary rehabilitation was associated with moderately lower short-term pain intensity and slightly lower disability than no rehabilitation (124). Moderate-quality evidence showed that multidisciplinary rehabilitation was associated with slightly lower short-term pain intensity and disability, moderately lower long-term pain intensity, and improved function compared with physical therapy and a greater likelihood of returning to work compared with nonmultidisciplinary rehabilitation (124).

Acupuncture

Low-quality evidence showed that acupuncture was associated with moderate improvement in pain relief immediately after treatment and up to 12 weeks later compared with sham acupuncture, but there was no improvement in function (125–130). Moderate-quality evidence showed that acupuncture was associated with moderately lower pain intensity and improved function compared with no acupuncture at the end of treatment (125). Low-quality evidence showed a small improvement in pain relief and function compared with medications (NSAIDs, muscle relaxants, or analgesics) (125).

Massage

Low-quality evidence showed no difference in pain between foot reflexology and usual care for patients with chronic low back pain (131–133). Moderate-quality evidence showed that massage improved short-term pain relief and function compared with other interventions (manipulation, exercise therapy, relaxation therapy, acupuncture, physiotherapy, or TENS) for patients with subacute to chronic low back pain, although effects were small (84, 86). Low-quality evidence showed that a combination of massage plus another intervention (exercise, exercise and education, or usual care) was superior to the other intervention alone for short-term pain in patients with subacute to chronic low back pain (84).

Spinal Manipulation

Low-quality evidence showed no difference in pain with spinal manipulation versus sham manipulation at 1 month (134, 135). Low-quality evidence showed that spinal manipulation slightly improved pain compared with an inert treatment (136–142). Moderate-quality evidence showed no clear differences in pain or function compared with another active intervention. Low-quality evidence showed that a combination of spinal manipulation with another active treatment resulted in greater pain relief and improved function at 1, 3, and 12

months compared with the other treatment alone (134, 143-147).

Ultrasound

Low-quality evidence showed no difference between ultrasound and sham ultrasound for pain at the end of treatment or 4 weeks after treatment (148-150). Low-quality evidence showed no difference between ultrasound and no ultrasound for pain or function (151, 152).

TENS

Low-quality evidence showed no difference between TENS and sham TENS for pain intensity or function at short-term follow-up (153). Low-quality evidence showed no difference between TENS and acupuncture in short- or long-term pain (154).

LLLT

Low-quality evidence showed that LLLT slightly improved pain compared with sham laser (155-157), and 1 RCT (155) showed that LLLT slightly improved function compared with sham laser.

Lumbar Support

Evidence was insufficient to compare lumbar support versus no lumbar support. Low-quality evidence showed no difference between a lumbar support plus exercise (muscle strengthening) versus exercise alone for pain or function at 8 weeks or 6 months (158). Low-quality evidence showed no clear differences between lumbar supports and other active treatments (traction, spinal manipulation, exercise, physiotherapy, or TENS) for pain or function (159-161).

Taping

Low-quality evidence showed no differences between Kinesio taping and sham taping for back-specific function after 5 or 12 weeks, although effects on pain were inconsistent between the 2 trials (162, 163). Low-quality evidence showed no differences between Kinesio taping and exercise for pain or function (164, 165).

Other Therapies

Evidence was insufficient to determine the effectiveness of electrical muscle stimulation, interferential therapy, short-wave diathermy, traction, or superficial heat or cold.

Radicular Low Back Pain

Exercise

Low-quality evidence showed that exercise resulted in small improvements in pain and function compared with usual care or no exercise (166-168).

Traction

Low-quality evidence showed no clear differences between traction and other active treatments, between

traction plus physiotherapy versus physiotherapy alone, or between different types of traction in patients with low back pain with or without radiculopathy (169).

Other Therapies

Evidence was insufficient for ultrasound, MCE, Pilates, tai chi, yoga, psychological therapies, multidisciplinary rehabilitation, acupuncture, massage, spinal manipulation, LLLT, electrical muscle stimulation, short-wave diathermy, TENS, interferential therapy, superficial heat or cold, lumbar support, and taping.

HARMS OF NONPHARMACOLOGIC THERAPIES

Evidence on adverse events from the included RCTs and systematic reviews was limited, and the quality of evidence for all available harms data is low. Harms were poorly reported (if they were reported at all) for most of the interventions.

Low-quality evidence showed no reported harms or serious adverse events associated with tai chi, psychological interventions, multidisciplinary rehabilitation, ultrasound, acupuncture, lumbar support, or traction (9, 95, 150, 170-174). Low-quality evidence showed that when harms were reported for exercise, they were often related to muscle soreness and increased pain, and no serious harms were reported. All reported harms associated with yoga were mild to moderate (119). Low-quality evidence showed that none of the RCTs reported any serious adverse events with massage, although 2 RCTs reported soreness during or after massage therapy (175, 176). Adverse events associated with spinal manipulation included muscle soreness or transient increases in pain (134). There were few adverse events reported and no clear differences between MCE and controls. Transcutaneous electrical nerve stimulation was associated with an increased risk for skin site reaction but not serious adverse events (177). Two RCTs (178, 179) showed an increased risk for skin flushing with heat compared with no heat or placebo, and no serious adverse events were reported. There were no data on cold therapy. Evidence was insufficient to determine harms of electrical muscle stimulation, LLLT, percutaneous electrical nerve stimulation, interferential therapy, short-wave diathermy, and taping.

COMPARISON OF CONCLUSIONS WITH THOSE OF THE 2007 GUIDELINE

Some evidence has changed since the 2007 ACP guideline and supporting evidence review. The 2007 review concluded that acetaminophen was effective for acute low back pain, based on indirect evidence from trials of acetaminophen for other conditions and trials of acetaminophen versus other analgesics. However, this update included a placebo-controlled RCT in patients with low back pain that showed no difference in effectiveness between acetaminophen and placebo (low-quality evidence). In addition, contrary to the 2007

review, current moderate-quality evidence showed that TCAs were not effective for chronic low back pain compared with placebo. Additional pharmacologic treatments addressed in the current review included duloxetine and the antiseizure medication pregabalin. Many conclusions about nonpharmacologic interventions are similar between the 2007 review and the update. Additional modalities assessed (with at least low-quality evidence) include mindfulness-based stress reduction, MCE, taping, and tai chi. Additional evidence or changes from the updated review include that superficial heat was found to be more effective for acute or subacute low back pain (moderate-quality evidence) and neither ultrasound nor TENS was shown to be effective compared with controls (low-quality evidence).

The **Figure** summarizes the recommendations and clinical considerations. Additional details on the evidence are available in **Appendix Tables 1 to 4** and the accompanying evidence reviews (7, 8).

RECOMMENDATIONS

Recommendation 1: Given that most patients with acute or subacute low back pain improve over time regardless of treatment, clinicians and patients should select nonpharmacologic treatment with superficial heat (moderate-quality evidence), massage, acupuncture, or spinal manipulation (low-quality evidence). If pharmacologic treatment is desired, clinicians and patients should select nonsteroidal anti-inflammatory drugs or skeletal muscle relaxants (moderate-quality evidence). (Grade: strong recommendation)

Clinicians should inform all patients of the generally favorable prognosis of acute low back pain with or without sciatica, including a high likelihood for substantial improvement in the first month (5, 180). Clinicians should also provide patients with evidence-based information with regard to their expected course, advise them to remain active as tolerated, and provide information about effective self-care options. Clinicians and patients should use a shared decision-making approach to select the most appropriate treatment based on patient preferences, availability, harms, and costs of the interventions. Nonpharmacologic interventions shown to be effective for improving pain and function in patients with acute or subacute low back pain include superficial heat (moderate-quality evidence and moderate improvement in pain and function) and massage (low-quality evidence and small to moderate improvement in pain and function). Low-quality evidence showed that acupuncture had a small effect on improving pain and spinal manipulation had a small effect on improving function compared with sham manipulation but not inert treatment. Harms of nonpharmacologic interventions were sparsely reported, and no serious adverse events were reported. Superficial heat was associated with increased risk for skin flushing, and massage and spinal manipulation were associated with muscle soreness.

We recommend that the choice between NSAIDs and SMRs be individualized on the basis of patient pref-

erences and likely individual medication risk profile. Treatment with NSAIDs resulted in a small improvement in both pain intensity (moderate-quality evidence) and function (low-quality evidence), and treatment with SMRs resulted in a small improvement in pain relief (moderate-quality evidence). There was no evidence for the effect of SMRs on function. Nonsteroidal anti-inflammatory drugs are associated with gastrointestinal and renal risks. Clinicians should therefore assess renovascular and gastrointestinal risk factors before prescribing NSAIDs and recommend the lowest effective doses for the shortest periods necessary. Although they are associated with lower risk for adverse effects than nonselective NSAIDs, COX-2-selective NSAIDs were not assessed for improvement in pain or function. Skeletal muscle relaxants are associated with central nervous system adverse effects, especially sedation.

The updated evidence showed that acetaminophen was not effective at improving pain outcomes versus placebo. However, this study assessed pain at 3 weeks after the intervention, and evidence from head-to-head trials showed no difference between acetaminophen and NSAIDs. Low-quality evidence showed that systemic steroids were not effective in treating acute or subacute low back pain, and we recommend against these drugs for treatment of acute low back pain.

Recommendation 2: For patients with chronic low back pain, clinicians and patients should initially select nonpharmacologic treatment with exercise, multidisciplinary rehabilitation, acupuncture, mindfulness-based stress reduction (moderate-quality evidence), tai chi, yoga, motor control exercise, progressive relaxation, electromyography biofeedback, low-level laser therapy, operant therapy, cognitive behavioral therapy, or spinal manipulation (low-quality evidence). (Grade: strong recommendation)

Nonpharmacologic interventions are considered as first-line options in patients with chronic low back pain because fewer harms are associated with these types of therapies than with pharmacologic options. It is important that physical therapies be administered by providers with appropriate training. Moderate-quality evidence showed that exercise therapy resulted in small improvements in pain and function. Specific components associated with greater effects on pain included individually designed programs, supervised home exercise, and group exercise; regimens that included stretching and strength training were most effective. Moderate-quality evidence showed that, compared with usual care, multidisciplinary rehabilitation resulted in moderate pain improvement in the short term (<3 months), small pain improvement in the long term, and small improvement in function in both the short and long term. Low-quality evidence showed that multidisciplinary rehabilitation resulted in a moderate improvement in pain and a small improvement in function compared with no multidisciplinary rehabilitation. Acupuncture had a moderate effect on pain and function compared with no acupuncture (moderate-quality evi-

Figure. Summary of the American College of Physicians guideline on noninvasive treatments for acute, subacute, or chronic low back pain.



Summary of the American College of Physicians Guideline on Noninvasive Treatments for Acute, Subacute, or Chronic Low Back Pain

Disease/Condition	Low back pain
Target Audience	All clinicians
Target Patient Population	Adults with acute, subacute, or chronic low back pain
Interventions Evaluated	Pharmacologic interventions: NSAIDs, nonopioid analgesics, opioid analgesics, tramadol and tapentadol, antidepressants, SMRs, benzodiazepines, corticosteroids, antiepileptic drugs Nonpharmacologic interventions: interdisciplinary or multicomponent rehabilitation; psychological therapies; exercise and related interventions, such as yoga or tai chi; complementary and alternative medicine therapies, including spinal manipulation, acupuncture, and massage; passive physical modalities, such as heat, cold, ultrasound, transcutaneous electrical nerve stimulation, electrical muscle stimulation, interferential therapy, short-wave diathermy, traction, LLLT, lumbar supports/braces
Outcomes Evaluated	Pain, function, health-related quality of life, work disability/return to work, global improvement, number of back pain episodes or time between episodes, patient satisfaction, adverse effects
Benefits	Acute low back pain Pharmacologic - NSAIDs: improved pain and function (small effect) - SMRs: improved pain (small effect) Nonpharmacologic - Heat wrap: improved pain and function (moderate effect) - Massage: improved pain and function (at 1 but not 5 wk) (small to moderate effect) - Acupuncture: improved pain (small effect) - Spinal manipulation: improved function (small effect) Chronic low back pain Pharmacologic - NSAIDs: improved pain (small to moderate effect) and function (no to small effect) - Opioids: improved pain and function (small effect) - Tramadol: improved pain (moderate effect) and function (small effect) - Buprenorphine (patch or sublingual): improved pain (small effect) - Duloxetine: improved pain and function (small effect) Nonpharmacologic - Exercise: improved pain and function (small effect) - Motor control exercise: improved pain (moderate effect) and function (small effect) - Tai chi: improved pain (moderate effect) and function (small effect) - Mindfulness-based stress reduction: improved pain and function (small effect) - Yoga: improved pain and function (small to moderate effect, depending on comparator) - Progressive relaxation: improved pain and function (moderate effect) - Multidisciplinary rehabilitation: improved pain (moderate effect) and function (no to small effect) - Acupuncture: improved pain (moderate effect) and function (no to moderate effect, depending on comparator) - LLLT: improved pain and function (small effect) - Electromyography biofeedback: improved pain (moderate effect) - Operant therapy: improved pain (small effect) - Cognitive behavioral therapy: improved pain (moderate effect) - Spinal manipulation: improved pain (small effect) Radicular low back pain - Exercise: improved pain or function (small effect)
Harms	Generally poorly reported Pharmacologic - NSAIDs: increased adverse effects compared with placebo and acetaminophen (COX-2–selective NSAIDs decreased risk for adverse effects compared with traditional NSAIDs) - Opioids: nausea, dizziness, constipation, vomiting, somnolence, and dry mouth - SMRs: increased risk for any adverse event and central nervous system adverse events (mostly sedation) - Benzodiazepines: somnolence, fatigue, lightheadedness - Antidepressants: increased risk for any adverse event Nonpharmacologic - Poorly reported, but no increase in serious adverse effects

Continued on following page

Figure—Continued

Recommendations	Recommendation 1: Given that most patients with acute or subacute low back pain improve over time regardless of treatment, clinicians and patients should select nonpharmacologic treatment with superficial heat (moderate-quality evidence), massage, acupuncture, or spinal manipulation (low-quality evidence). If pharmacologic treatment is desired, clinicians and patients should select nonsteroidal anti-inflammatory drugs or skeletal muscle relaxants (moderate-quality evidence). (Grade: strong recommendation) Recommendation 2: For patients with chronic low back pain, clinicians and patients should initially select nonpharmacologic treatment with exercise, multidisciplinary rehabilitation, acupuncture, mindfulness-based stress reduction (moderate-quality evidence), tai chi, yoga, motor control exercise, progressive relaxation, electromyography biofeedback, low-level laser therapy, operant therapy, cognitive behavioral therapy, or spinal manipulation (low-quality evidence). (Grade: strong recommendation) Recommendation 3: In patients with chronic low back pain who have had an inadequate response to nonpharmacologic therapy, clinicians and patients should consider pharmacologic treatment with nonsteroidal anti-inflammatory drugs as first-line therapy, or tramadol or duloxetine as second-line therapy. Clinicians should only consider opioids as an option in patients who have failed the aforementioned treatments and only if the potential benefits outweigh the risks for individual patients and after a discussion of known risks and realistic benefits with patients. (Grade: weak recommendation, moderate-quality evidence)
High-Value Care	Clinicians should reassure patients that acute or subacute low back pain usually improves over time regardless of treatment and should avoid prescribing costly and potentially harmful treatments. Systemic steroids were not shown to provide benefit and should not be prescribed for patients with acute or subacute low back pain, even with radicular symptoms. For treatment of chronic low back pain, clinicians should select therapies that have the fewest harms and lowest costs. Clinicians should avoid prescribing costly therapies and those with substantial potential harms, such as long-term opioids, and pharmacologic therapies that were not shown to be effective, such as tricyclic antidepressants and selective serotonin reuptake inhibitors.
Clinical Considerations	Clinicians should inform patients with acute or subacute low back pain of the generally very favorable outcome. Thus, patients can avoid potentially harmful and costly tests and treatments. Clinicians should advise patients with acute, subacute, or chronic low back pain to remain active as tolerated. Improvements in pain and function due to pharmacologic and nonpharmacologic interventions were small and often showed no clear differences compared with controls. Few differences in recommended therapies were found when they were studied in head-to-head trials. Therefore, clinicians should base treatment recommendations on patient preferences that also minimize harms and costs.

COX-2 = cyclooxygenase-2; LLLT = low-level laser therapy; NSAID = nonsteroidal anti-inflammatory drug; SMR = skeletal muscle relaxant.

dence) and a moderate effect on pain with no clear effect on function compared with sham acupuncture (low-quality evidence). Moderate-quality evidence showed that mindfulness-based stress reduction resulted in small improvements in pain and function (small effect), and 1 study showed that it was equivalent to CBT for improving back pain and function.

Low-quality evidence showed that tai chi had a moderate effect on pain and a small effect on function. Tai chi sessions in included studies lasted 40 to 45 minutes and were done 2 to 5 times per week for 10 to 24 weeks. Low-quality evidence showed that yoga improved pain and function by a moderate amount compared with usual care and by a small amount compared with education. Low-quality evidence showed that MCE had a moderate effect on pain and a small effect on function. Motor control exercise, tai chi, and yoga were favored over general exercise (low-quality evidence).

Low-quality evidence showed that progressive relaxation had a moderate effect on pain and function, electromyography biofeedback and CBT each had a moderate effect on pain and no effect on function, and operant therapy had a small effect on pain and no effect on function. Low-quality evidence showed that LLLT had a small effect on pain and function. Low-quality evidence showed that spinal manipulation had a small effect on pain compared with inert treatment but no effect compared with sham manipulation. There

were no clear differences between spinal manipulation and other active interventions (moderate-quality evidence).

Harms were poorly reported for nonpharmacologic therapies, although no serious harms were reported for any of the recommended interventions. Muscle soreness was reported for exercise, massage, and spinal manipulation.

Ultrasound, TENS, and Kinesio taping had no effect on pain or function compared with control treatments (low-quality evidence).

Recommendation 3: In patients with chronic low back pain who have had an inadequate response to nonpharmacologic therapy, clinicians and patients should consider pharmacologic treatment with nonsteroidal anti-inflammatory drugs as first-line therapy, or tramadol or duloxetine as second-line therapy. Clinicians should only consider opioids as an option in patients who have failed the aforementioned treatments and only if the potential benefits outweigh the risks for individual patients and after a discussion of known risks and realistic benefits with patients. (Grade: weak recommendation, moderate-quality evidence)

Pharmacologic therapy should be considered for patients with chronic low back pain who do not improve with nonpharmacologic interventions. Nonsteroidal anti-inflammatory drugs had a small to moderate effect on pain (moderate-quality evidence) and no to

small effect on function (low-quality evidence) and should be the first option considered. Moderate-quality evidence showed no difference in pain improvement when different NSAIDs were compared with one another. Nonsteroidal anti-inflammatory drugs are associated with gastrointestinal and renal risks. Clinicians should therefore assess renovascular and gastrointestinal risk factors before prescribing NSAIDs and should recommend the lowest effective doses for the shortest periods necessary. COX-2-selective NSAIDs were not assessed for improvement in pain or function, although they are associated with lower risk for adverse effects than nonselective NSAIDs.

For second-line therapies, moderate-quality evidence showed that tramadol had a moderate effect on pain and a small effect on function in the short term. Of note, tramadol is a narcotic and, like other opioids, is associated with the risk for abuse (181). Moderate-quality evidence showed that duloxetine had a small effect on pain and function.

Moderate-quality evidence showed that opioids (morphine, oxycodone, hydromorphone, and tapentadol) had a small effect on short-term pain and function. Low-quality evidence showed that buprenorphine (patch or sublingual) resulted in a small improvement in pain. Opioids should be the last treatment option considered and should be considered only in patients for whom other therapies have failed because they are associated with substantial harms. Moderate-quality evidence showed no difference in pain or function when different long-acting opioids were compared with one another. Harms of short-term use of opioids include increased nausea, dizziness, constipation, vomiting, somnolence, and dry mouth compared with placebo. Studies assessing opioids for the treatment of chronic low back pain did not address the risk for addiction, abuse, or overdose, although observational studies have shown a dose-dependent relationship between opioid use for chronic pain and serious harms (182).

Moderate-quality evidence showed that TCAs did not effectively improve pain or function (low-quality evidence) in patients with chronic low back pain, which is contrary to the 2007 guideline. In addition, moderate-quality evidence showed that SSRIs did not improve pain.

AREAS OF INCONCLUSIVE EVIDENCE

Evidence is insufficient or lacking to determine treatments for radicular low back pain. Most RCTs enrolled a mixture of patients with acute, subacute, and chronic low back pain, so it is difficult to extrapolate the benefits of treatment compared with its duration. Use of opioids for chronic pain is an important area that requires further research to compare benefits and harms of therapy. The evidence is also insufficient for most physical modalities. Evidence is insufficient on which patients are likely to benefit from which specific therapy. Evidence on patient-important outcomes, such as disability or return to work, was largely unavailable,

and available evidence showed no clear connection with improvements in pain.

HIGH-VALUE CARE

Clinicians should reassure patients that acute or subacute low back pain usually improves over time, regardless of treatment. Thus, clinicians should avoid prescribing costly and potentially harmful treatments for these patients, especially narcotics. In addition, systemic steroids were not shown to provide benefit and should not be prescribed for patients with acute or subacute low back pain, even with radicular symptoms. For treatment of chronic low back pain, clinicians should select therapies that have the fewest harms and lowest costs because there were no clear comparative advantages for most treatments compared with one another. Clinicians should avoid prescribing costly therapies; those with substantial potential harms, such as long-term opioids (which can be associated with addiction and accidental overdose); and pharmacologic therapies that were not shown to be effective, such as TCAs and SSRIs.

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Note: Clinical practice guidelines are “guides” only and may not apply to all patients and all clinical situations. Thus, they are not intended to override clinicians’ judgment. All ACP clinical practice guidelines are considered automatically withdrawn or invalid 5 years after publication or once an update has been issued.

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APPENDIX: DETAILED METHODS

The evidence review was conducted by the AHRQ's Pacific Northwest Evidence-based Practice Center. Details of the ACP guideline development process can be found in ACP's methods paper (12). Disclosures of interests and management of any conflicts can be found at www.acponline.org/clinical_information/guidelines/guidelines/conflicts_cgc.htm.

Key Questions Addressed

1. What are the comparative benefits and harms of different pharmacologic therapies for acute or chronic nonradicular low back pain, radicular low back pain, or spinal stenosis, including NSAIDs, acetaminophen, opioids, muscle relaxants, antiseizure medications, antidepressants, corticosteroids, and topical or patch-delivered medications?

2. What are the comparative benefits and harms of different nonpharmacologic, noninvasive therapies for acute or chronic nonradicular low back pain, radicular low back pain, or spinal stenosis, including but not limited to interdisciplinary rehabilitation, exercise (various types), physical modalities (ultrasound, TENS, electrical muscle stimulation, interferential therapy, heat [various forms], and ice), traction tables/devices, back supports/bracing, spinal manipulation, various psychological therapies, acupuncture, massage therapy (various types), yoga, magnets, and low-level lasers?

Search Strategy

Reviewers searched MEDLINE, the Cochrane Central Register of Controlled Trials, and the Cochrane Database of Systematic Reviews for trials published from January 2008 through April 2015. Searches were updated through November 2016. Studies published before 2008 were identified using the 2007 ACP/APS systematic reviews (10, 11).

Quality Assessment

Randomized trials were evaluated using methods developed by the Cochrane Back Review Group and the AHRQ (183), and systematic reviews were assessed using AMSTAR (A Measurement Tool to Assess Systematic Reviews) (184).

Population Studied

Adults with acute, subacute, or chronic nonradicular low back pain, radicular low back pain, or symptomatic spinal stenosis.

Interventions Evaluated

Oral or topical pharmacologic therapies included NSAIDs, acetaminophen, opioids, tramadol and tapentadol, antidepressants, SMRs, benzodiazepines, corticosteroids, antiepileptic medications, capsaicin, and lidocaine.

Noninvasive, nonpharmacologic therapies included interdisciplinary or multicomponent rehabilitation (physical therapy plus psychological therapy with some coordination), psychological therapies, exercise and related interventions (such as yoga or tai chi), complementary and alternative medicine therapies (spinal manipulation, acupuncture, and massage), passive physical modalities (such as heat, cold, ultrasound, TENS, electrical muscle stimulation, interferential therapy, short-wave diathermy, traction, LLLT, and lumbar supports/braces), and taping.

Comparators

Interventions were compared with each other or with placebo (drug trials), sham (functionally inert) treatments, or no treatment.

Outcomes

Outcomes included reduction or elimination of low back pain (including related leg symptoms), improvement in back-specific and overall function, improvement in health-related quality of life, reduction in work disability and return to work, global improvement, number of back pain episodes or time between episodes, patient satisfaction, and adverse effects of interventions.

Timing

Timing of outcomes was stratified as long-term (≥ 1 year) and short-term (≤ 6 months).

Setting

Settings included inpatient and outpatient.

Target Audience

The target audience includes all clinicians.

Target Patient Population

The target patient population includes adults with acute (<4 weeks), subacute (4 to 12 weeks), or chronic (>12 weeks) nonradicular low back pain, radicular low back pain, or symptomatic spinal stenosis. Children or adolescents with low back pain; pregnant women; and patients with low back pain from sources outside the back (nonspinal low back pain), fibromyalgia or other myofascial pain syndromes, and thoracic or cervical back pain are not included.

Peer Review

The AHRQ systematic review was sent to invited peer reviewers and posted on the AHRQ Web site for

public comments. The accompanying evidence reviews (7, 8) also underwent a peer review process through the journal. The guideline underwent a peer review process through the journal and was posted online for comments from ACP Regents and ACP Governors, who represent ACP members at the regional level.

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Appendix Table 1. Pharmacologic and Nonpharmacologic Treatments for Acute or Subacute Low Back Pain

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Pharmacologic treatments vs. placebo (acute only)				
Acetaminophen Pain Function NSAIDs	Pain	No effect	Low (1 RCT)	0 to 10 scale: Score differences, ≤ 0.20 point
	Function	No effect	Low (1 RCT)	RDQ: Score differences, ≤ 0.60 point
	Pain	Small (pain intensity) No effect (pain relief)	Moderate (5 RCTs)	0 to 100 scale: WMD, -8.39 (95% CI, -12.68 to -4.10 ; chi-square, 3.47; $P > 0.10$)
	Function SMRs	Small	Low (2 RCTs)	0 to 24 RDQ: Score differences, 2.4 to 2.9 points; $P < 0.001$
Systemic corticosteroids Pain Function	Pain	Small	Moderate (5 RCTs)	0- to 10-point visual analogue scale 2 to 4 d: RR, 1.25 (CI, 1.12 to 1.41) 5 to 7 d: RR, 1.72 (CI, 1.32 to 2.22)
	Function	No effect	Low (2 RCTs)	No clear difference (single intramuscular injection or a 5-d course of systemic corticosteroids)
	Pain	No effect	Low (2 RCTs)	
	Function	No effect	Low (2 RCTs)	
Nonpharmacologic treatments vs. sham, no treatment, or usual care (acute or subacute)				
Exercise vs. usual care Pain Function	Pain	No effect	Low (6 RCTs)	0 to 100 scale Acute, intermediate-term: WMD, 0.59 (CI, -11.51 to 12.69) Subacute: WMD, 1.89 (CI, -1.13 to 4.91)
	Function	No effect	Low (6 RCTs)	Acute, short-term: WMD, -2.82 (CI, -15.35 to 9.71) Acute, intermediate-term: WMD, 2.47 (CI, -0.26 to 5.21) Subacute: WMD, 1.07 (CI, -3.18 to 5.32)
Acupuncture vs. sham acupuncture Pain Function	Pain	Small	Low (2 RCTs)	0 to 100 scale MD, 9.38 (CI, 1.76 to 17.0; $I^2 = 27\%$) 3 other trials reported effects consistent with these findings
	Function	No effect	Low (5 RCTs)	No clear effect
Massage vs. sham massage Pain Function	Pain	1 wk: Moderate 5 wk: No effect	Low (2 RCTs)	1 wk: SMD, -0.92 (CI, -1.35 to -0.48) There was no significant difference in pain at 5 wk in 1 trial
	Function	1 wk: Moderate 5 wk: No effect	Low (2 RCTs)	1 wk: SMD, -1.76 (CI, -3.19 to -0.32) There was no significant difference in function at 5 wk in 1 trial
Spinal manipulation vs. inert treatment Pain Function	Pain	No effect	Low (3 RCTs)	0 to 10 scale at 1 wk: WMD, 0.14 (CI, -0.69 to 0.96 ; $I^2 = 27\%$), although 1 trial found spinal manipulation to be associated with better pain relief at 3 mo: MD, -1.20 (CI, 2.11 to -0.29)
	Function	No effect	Low (2 RCTs)	1 wk: SMD, -0.08 (CI, -0.37 to 0.21 ; $I^2 = 0\%$) 3 mo: SMD, -0.28 (CI, -0.59 to 0.02)
Spinal manipulation vs. sham treatment Function Heat wrap vs. placebo Pain Function	Function	Small	Low (2 RCTs)	Statistically significant in 1 trial
	Pain	Moderate	Moderate (4 RCTs)	0 to 5 scale, 5 d: MD, 1.06 (CI, 0.68 to 1.45)
	Function	Moderate	Moderate (2 RCTs)	0 to 100 scale, 3 to 4 d: score differences, 16 to 20 points RDQ, 4 d: MD, -2.10 (CI, -3.19 to -1.01)

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Appendix Table 1—Continued

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Comparative benefits of pharmacologic and nonpharmacologic treatments				
NSAID vs. NSAID				
Pain		No difference	Moderate (21 RCTs)	No reported differences in 15 of 21 trials
Acetaminophen vs. NSAID				
Pain		No difference	Low (4 RCTs)	Pooled SMD, 0.21 (CI, -0.02 to 0.43)
Function		No difference	Low (4 RCTs)	RR, 0.81 (CI, 0.58 to 1.14)
COX-2-selective NSAID vs. traditional NSAID				
Pain		No difference	Low (3 RCTs)	No clear differences
SMR vs. another SMR				
Pain		No difference	Low (2 RCTs)	No differences for carisoprodol vs. cyclobenzaprine or
Function		No difference	Low (2 RCTs)	tizanidine vs. chlorzoxazone
Exercise vs. exercise				
Pain		No difference	Moderate (>20 RCTs)	No clear differences in >20 head-to-head trials of patients
Function		No difference	Moderate (>20 RCTs)	
Lumbar support vs. other active treatments				
Pain		No difference	Low (3 RCTs)	3 trials found no clear differences between lumbar supports
Function		No difference	Low (3 RCTs)	vs. other active treatments in pain or function
Acupuncture vs. NSAIDs				
Overall improvement		Small	Low (5 RCTs)	RR, 1.11 (CI, 1.06 to 1.16; $I^2 = 0\%$)
Spinal manipulation vs. other active treatments (exercise, physical therapy, or back school)				
Pain		No difference	Moderate (3 RCTs)	0 to 10 scale 1 wk: WMD (CI, -0.53 to 0.65; $I^2 = 0\%$) 1 mo: WMD, -0.15 (CI, -0.49 to 0.18; $I^2 = 0\%$) 3 to 6 mo: WMD, -0.20 (CI, -1.13 to 0.73; $I^2 = 81\%$) 1 y: mean difference, 0.40 (CI, -0.08 to 0.88) Findings were similar for function, with no differences observed at any time point
Function		No difference	Moderate (3 RCTs)	
Heat vs. simple analgesics				
Pain		Small	Low (1 RCT)	0 to 10 scale 1 to 2 d, acetaminophen: MD, 0.90 (CI, 0.50 to 1.30) 1 to 2 d, ibuprofen: MD, 0.65 (CI, 0.25 to 1.05)
Function		Small	Low (1 RCT)	RDQ 1 to 2 d, acetaminophen: MD, 2.00 (CI, 0.86 to 3.14) 1 to 2 d, ibuprofen: MD, 2.20 (CI, 1.11 to 3.29)
Heat vs. exercise				
Pain		No difference	Low (1 RCT)	0 to 10 scale Days 1 to 2: MD, 0.40 (CI, -0.15 to 0.95) Day 7: MD, 0.30 (CI, -0.68 to 1.28)
Function		No difference	Low (1 RCT)	RDQ Day 4: MD, -0.70 (CI, -2.09 to 0.69) Day 7: MD, -0.90 (CI, -2.84 to 1.04)

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Appendix Table 1—Continued

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Combination therapy treatments vs. monotherapy treatments or no treatment (acute or subacute)				
SMR + NSAID vs. NSAID alone				
Pain		Inconsistent benefit	Low (3 RCTs)	Estimate from 3 trials favored combination therapy: RR, 1.56 (CI, 0.92 to 2.70; $I^2 = 84\%$), but another trial found no effect
Massage vs. other treatments (manipulation, exercise therapy, relaxation therapy, acupuncture, or physiotherapy) vs. other treatment alone (subacute to chronic)				
Pain		Small	Moderate (9 RCTs)	0 to 10 scale, short-term: Better effects in 7 of 9 trials (MDs, -0.6 to -0.94 points)
Function		Small	Moderate (4 RCTs)	Short-term: Better effects on short-term function in 3 of 4 trials
Massage plus another active treatment (exercise, exercise and education, or usual care) vs. the other treatment alone (subacute to chronic)				
Pain		Improved	Low (5 RCTs)	Short-term: Superior to other intervention without massage; effects stronger when massage combined with exercise
Function		Improved	Low (5 RCTs)	Long-term: Few differences
Spinal manipulation plus exercise or advice vs. exercise or advice alone				
Function		1 wk: Small 1 or 3 mo: No difference	Low (4 RCTs)	1 wk: SMD, -0.41 (CI, -0.73 to -0.10 ; $I^2 = 18\%$) 1 mo: SMD, -0.09 (CI, -0.39 to 0.21 ; $I^2 = 37\%$) 3 mo: SMD, -0.22 (CI, -0.61 to 0.16 ; $I^2 = 41\%$)
Heat plus exercise vs. exercise alone				
Pain		Small	Low (1 RCT)	0 to 10 scale: MD, 1.40 (CI, 0.69 to 2.11)
Function		Small	Low (1 RCT)	RDQ at 7 d: MD, -3.20 (CI, -5.42 to 0)
LLLT + NSAID vs. sham + NSAID				
Pain		Large	Low (1 RCT)	0 to 100 scale at 3 wk: Mean change, -30.0 vs. -15.7 vs. -20.8
Function		Moderate	Low (1 RCT)	ODI at 3 wk: Mean change, -12.0 vs. -6.5 vs. -10.0
Lumbar support plus education vs. education				
Pain		No difference	Low (1 RCT)	No reported differences after 1 y
Function		No difference	Low (1 RCT)	

COX-2 = cyclooxygenase-2; LLLT = low-level laser therapy; MD = mean difference; NSAID = nonsteroidal anti-inflammatory drug; ODI = Oswestry Disability Index; RCT = randomized, controlled trial; RDQ = Roland Morris Disability Questionnaire; RR = relative risk; SMD = standardized mean difference; SMR = skeletal muscle relaxant; WMD = weighted mean difference.

Appendix Table 2. Pharmacologic and Nonpharmacologic Treatments for Chronic Low Back Pain

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Pharmacologic treatments vs. placebo				
NSAIDs				
Pain	Small to moderate		Moderate (6 RCTs)	0 to 100 scale, 12 wk: WMD, -12.40 (95% CI, -15.53 to -9.26; chi-square, 1.82; $P > 0.5$) 0 to 10 scale, 12 to 16 wk: Score changes, 0.41 to 0.59 ≥30% pain relief: 56.8% vs. 31.7% and 37% vs. 27% RDC: MDs, ~0.02 to 2.00 points
Function	Small to no effect		Low (4 RCTs)	
Strong opioids				
Pain	Small		Moderate (10 RCTs)	0 to 10 scale: SMD, -0.43 (CI, -0.52 to -0.33; $I^2 = 0\%$; MD, ~1 point)
Function	Small		Moderate (8 RCTs)	RDC: SMD, -0.26 (CI, -0.37 to -0.15; $I^2 = 0\%$; MD, ~1 point)
Tramadol				
Pain	Moderate		Moderate (7 RCTs)	0 to 10 scale: SMD, -0.55 (CI, -0.66 to -0.44; $I^2 = 86\%$; MD, ≤1 point)
Function	Small		Moderate (7 RCTs)	RDC: SMD, -0.18 (CI, -0.29 to -0.07; $I^2 = 0\%$; MD, ~1 point)
Opioids: buprenorphine patch or sublingual (subacute or chronic)				
Pain	Small		Low (3 RCTs)	0 to 10 scale: Score difference, ~1 point
Tetrazepam				
Pain	Lower likelihood of failure to improve		Low (2 RCTs)	5 to 7 d: RR, 0.82 (CI, 0.72 to 0.94) 10 to 14 d: RR, 0.71 (CI, 0.54 to 0.93)
Overall improvement	Lower likelihood of failure to improve		Low (2 RCTs)	10 to 14 d: RR, 0.63 (CI, 0.42 to 0.97)
TCA				
Pain	No effect		Moderate (4 RCTs)	SMD, -0.10 (CI, -0.51 to 0.31; $I^2 = 32\%$)
Antidepressants				
Function	No effect		Low (2 RCTs)	SMD, -0.06 (CI, -0.40 to 0.29; $I^2 = 0\%$)
SSRI				
Pain	No effect		Moderate (3 RCTs)	SMD, 0.11 (CI, -0.17 to 0.39; $I^2 = 0\%$)
Duloxetine				
Pain	Small		Moderate (3 RCTs)	0 to 10 scale: Mean between-group difference, 0.58 to 0.74
Function	Small		Moderate (3 RCTs)	RDC: Mean change from baseline, -2.69 vs. -2.22; $P = 0.26$
Nonpharmacologic treatments vs. sham, no treatment, or usual care				
Exercise vs. no exercise				
Pain	Small		Moderate (19 RCTs)	0 to 100 scale: WMD, 10.0 (CI, 1.31 to 19.09)
Function	Small		Moderate (18 RCTs)	0 to 100 scale: Not statistically significant; WMD, 3.0 (CI, -0.53 to 6.48)
Exercise vs. usual care				
Pain	Small		Moderate (3 RCTs)	0 to 100 scale Treatment end: WMD, -9.23 (CI, -16.02 to -2.43) Long-term: MD, -4.94 (CI, -10.45 to 0.58)
Function	Small		Moderate (18 RCTs)	0 to 100 scale Treatment end: WMD, -12.35 (CI, -23.0 to -1.69) Long-term: MD, -3.17 (CI, -5.96 to -0.38)
MCE vs. minimal intervention				
Pain	Moderate		Low (2 RCTs)	0 to 100 scale Short-term: WMD, -12.48 (CI, -19.04 to -5.93) Intermediate-term: WMD, -10.18 (CI, -16.64 to -3.72) Long-term: WMD, -13.32 (CI, -19.75 to -6.90)
Function	Small		Low (3 RCTs)	0 to 100 scale Short-term: WMD, -9.00 (CI, -15.28 to -2.73) Intermediate-term: WMD, -5.62 (CI, -10.46 to -0.77) Long-term: WMD, -6.64 (CI, -11.72 to -1.57)
Tai chi vs. wait list or no tai chi				
Pain	Moderate		Low (2 RCTs)	0 to 10 scale: MDs, 0.9 and 1.3
Function	Small		Low (1 RCT)	RDC: MD, 2.6 (CI, 1.1 to 3.7)

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Appendix Table 2–Continued

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Yoga vs. usual care				
Pain		Moderate	Low (1 RCT)	0 to 100 VAS, 24 wk: Mean scores, 24 vs. 37 ($P < 0.001$)
Function		Moderate	Low (1 RCT)	0 to 100 ODI, 24 wk: Mean scores, 18 vs. 21 ($P < 0.01$)
Yoga vs. education				
Pain		Short-term: Small Long-term: No difference	Low (5 RCTs)	Short-term: SMD, -0.45 (CI, -0.63 to -0.26 ; $I^2 = 0\%$) Long-term: Not statistically significant; SMD, -0.28 (CI, -0.58 to -0.02 ; $I^2 = 47\%$)
Function		Small	Low (5 RCTs)	Short-term: SMD, 0.45 (CI, -0.65 to -0.25 ; $I^2 = 8\%$) Long-term: SMD, 0.39 (CI, -0.66 to -0.11 ; $I^2 = 40\%$)
Mindfulness-based stress reduction vs. usual care				
Pain		Improved	Moderate (3 RCTs)	0 to 10 scale, 26 wk: Score difference, -0.64 $\geq 30\%$ improvement: RR, 1.64 (CI, 1.15 to 2.34)
Function		Improved	Moderate (3 RCTs)	RDO, 26 wk: score difference, -1.37 $\geq 30\%$ improvement: RR, 1.37 (CI, 1.06 to 1.77)
Progressive relaxation vs. wait-list control				
Pain		Moderate	Low (3 RCTs)	0 to 100 VAS: MD, -19.77 (CI, -34.0 to -5.20 ; $I^2 = 57\%$)
Function		Moderate	Low (3 RCTs)	SMD, -0.88 (CI, -1.36 to -0.39 ; $I^2 = 0\%$)
Electromyography biofeedback vs. wait-list control or placebo				
Pain		Moderate	Low (3 RCTs)	SMD, -0.80 (CI, -1.32 to -0.28 ; $I^2 = 0\%$)
Function		No effect	Low (3 RCTs)	No clear effect
Operant therapy vs. wait-list control				
Pain		Small	Low (3 RCTs)	0 to 100 VAS or 0 to 78 McGill: SMD, -0.43 (CI, -0.75 to -0.11 ; $I^2 = 0\%$)
Function		No effect	Low (2 RCTs)	0 to 100 Sickness Impact Profile: MD, -1.18 (CI, -3.53 to 1.18)
CBT vs. wait-list control				
Pain		Moderate	Low (5 RCTs)	0 to 100 VAS or 0 to 78 McGill: SMD, -0.60 (CI, -0.97 to -0.22 ; $I^2 = 40\%$)
Function		No effect	Low (4 RCTs)	Sickness Impact Profile: Not statistically significant; SMD, -0.37 (CI, -0.87 to 0.13 ; $I^2 = 50\%$)
Multidisciplinary rehabilitation vs. usual care				
Pain		Short-term: Moderate Long-term: Small	Moderate (9 RCTs)	0 to 10 scale <3 mo: SMD, -0.55 (CI, -0.83 to -0.28) or ~ 1.4 -point MD Long-term: SMD, -0.21 (CI, -0.37 to -0.04) or ~ 0.5 -point MD
Disability		Small	Moderate (9 RCTs)	RDO <3 mo: SMD, -0.41 (CI, -0.62 to -0.19) or ~ 2.5 -point MD Long-term: SMD, -0.23 (CI, -0.40 to -0.06) or ~ 1.4 -point MD
Return to work		No effect	Moderate (7 RCTs)	Short-term: OR, 1.07 (CI, 0.60 to 1.90) Long-term: OR, 1.04 (CI, 0.73 to 1.47)
Multidisciplinary rehabilitation vs. no multidisciplinary rehabilitation				
Pain		Moderate	Low (3 RCTs)	0 to 10 scale: SMD, -0.73 (CI, -1.22 to -0.24 ; $I^2 = 64\%$) or ~ 1.7 -point MD
Disability		Small	Low (3 RCTs)	RDO: Pooled SMD, -0.49 (CI, -0.76 to -0.22 ; $I^2 = 0\%$) or ~ 2.9 -point MD
Acupuncture vs. sham acupuncture		Moderate	Low (9 RCTs)	Immediately at the end of treatment: WMD, -16.76 (CI, -33.3 to -0.19 ; $I^2 = 90\%$)
Function		No effect	Low (9 RCTs)	12 wk: WMD, -9.55 (CI, -16.5 to -2.58 ; $I^2 = 40\%$) No reported differences
Acupuncture vs. no acupuncture				
Pain		Moderate	Moderate (4 RCTs)	Immediately: SMD, -0.72 (CI, -0.94 to -0.49 ; $I^2 = 51\%$)
Function		Moderate	Moderate (3 RCTs)	Immediately: SMD, -0.94 (CI, -1.41 to -0.47 ; $I^2 = 78\%$)

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Appendix Table 2–Continued

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Massage vs. usual care Pain		No effect	Low (1 RCT)	1 trial found no difference between foot reflexology vs. usual care in pain or function, and 1 trial found structural or relaxation massage to be associated with better function (mean, 2.5 to 2.9 points on the RDQ) vs. usual care at 10 wk; effects were less pronounced at 52 wk
Spinal manipulation vs. sham treatment Pain		No effect	Low (4 RCTs)	0 to 100 scale, 1 mo: WMD, -3.24 (CI, -13.62 to 7.15)
Spinal manipulation vs. inert treatment Pain		Small	Low (7 RCTs)	0 to 10 scale: MD, 0.9 (CI, 0.1 to 1.7)
Ultrasound vs. sham ultrasound Pain		No effect	Low (5 RCTs)	0 to 100 scale Immediately: MD, -7.12 (CI, -18.0 to 3.75 ; $I^2 = 77\%$) 4 wk: No reported differences in 2 trials
Ultrasound vs. no ultrasound Pain		No effect	Low (5 RCTs)	0 to 100 scale: MD, -2.16 (CI, -4.66 to 0.34 ; $I^2 = 0\%$)
Function		No effect	Low (5 RCTs)	0 to 100 scale: MD, -0.41 (CI, -3.14 to 2.32)
TENS vs. sham treatment Pain		No effect	Low (4 RCTs)	0 to 100 scale: WMD, -4.47 (CI, -12.84 to 3.89)
Disability		No effect	Low (2 RCTs)	0 to 100 scale: WMD, -1.36 (CI, -4.38 to 1.66)
LLLT vs. sham laser Pain		Small	Low (3 RCTs)	3 of 4 trials showed improvement
Function		Small	Low (1 RCT)	1 trial showed improvement
Kinesio taping vs. sham taping Function		No effect	Low (2 RCTs)	No effect on back-specific function at 5 or 12 wk
Comparative benefits of pharmacologic and nonpharmacologic treatments				
NSAID vs. NSAID Pain		No difference	Moderate (6 RCTs)	No reported differences in 6 of 6 trials
Long-acting opioids vs. long-acting opioids Pain		No difference	Moderate (4 RCTs)	No clear differences (oral morphine vs. transdermal fentanyl or oxycodone vs. oxycodone or morphine vs. oxycodone)
Function		No difference	Moderate (4 RCTs)	
Long-acting opioids vs. short-acting opioids* Pain		No difference	Low (6 RCTs)	No clear differences
SMR vs. another SMR Pain		No difference	Low (2 RCTs)	No reported differences (pridinol vs. thiocolchicoside)
Diazepam vs. cyclobenzaprine Muscle spasms		No difference	Low (2 RCTs)	No clear difference
MCE vs. general exercise Pain		Small	Low (6 RCTs)	0 to 100 scale Short-term: WMD, -7.80 (CI, -10.95 to -4.65) Intermediate-term: WMD, -6.06 (CI, -10.94 to -1.18) Long-term: Not statistically significant; WMD, -3.10 (CI, -7.03 to 0.83)
Function		Small	Low (6 RCTs)	0 to 100 scale Short-term: WMD, -4.65 (CI, -6.20 to -3.11) Long-term: WMD, -4.72 (CI, -8.81 to -0.63)
MCE vs. multimodal physical therapy Pain		Moderate	Low (4 RCTs)	0 to 100 scale, intermediate-term: WMD, -14.20 (CI, -21.23 to -7.16)
Function		Moderate	Low (2 RCTs)	0 to 100 scale, intermediate-term: WMD, -12.98 (CI, -19.49 to -6.47)

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Appendix Table 2—Continued

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Exercise vs. exercise				
Pain		No difference	Moderate (>20 RCTs)	No clear differences in >20 head-to-head trials of patients
Function		No difference	Moderate (>20 RCTs)	
Pilates vs. usual care + physical activity				
Pain		Small to no effect	Low (7 RCTs)	Small (MD, -1.6 to -4.1 points) to no effect on pain
Function		No effect	Low (7 RCTs)	No clear effects
Pilates vs. other exercise				
Pain		No difference	Low (3 RCTs)	No clear differences
Function		No difference	Low (3 RCTs)	
Tai chi vs. other exercise				
Pain		Moderate	Low (1 RCT)	Backward walking or jogging through 6 mo: MDs, -0.7 and -0.8 Swimming: No reported differences (MD, -0.1 at 3 and 6 mo)
Yoga vs. exercise				
Pain		Small	Low (5 RCTs)	Lower pain intensity vs. exercise in most trials, although effects were small and differences were not always statistically significant
Psychological therapies vs. exercise or physical therapy				
Pain		No difference	Low (6 RCTs)	No clear differences
Psychological therapies vs. other psychological therapies				
Pain		No difference	Moderate (10 RCTs)	No clear differences
Function		No difference	Moderate (10 RCTs)	
Multidisciplinary rehabilitation vs. physical therapy				
Pain		Short-term: Small Long-term: Moderate	Moderate (13 RCTs)	0 to 10 NRS Short-term: SMD, -0.30 (CI, -0.54 to -0.06) or ~0.6-point MD Long-term: SMD, -0.51 (CI, -1.04 to 0.01) or ~1.2-point MD
Function		Short-term: Small Long-term: Moderate	Moderate (13 RCTs)	0 to 10 NRS, short-term: SMD, -0.39 (CI, -0.68 to -0.10) or ~1.2-point MD RDQ, long-term function: SMD, -0.68 (CI, -1.19 to -0.16) or ~4.0-point MD Greater likelihood of return to work: OR, 1.87 (CI, 1.39 to 2.53)
Acupuncture vs. medications (NSAIDs, muscle relaxants, and analgesics)				
Pain		Small	Low (3 RCTs)	0 to 100 scale, immediately: WMD, -10.56 (CI, -20.34 to -0.78)
Function		Small	Low (3 RCTs)	0 to 100 scale, immediately: SMD, -0.36 (CI, -0.67 to -0.04)
Spinal manipulation vs. other treatments (exercise, usual care, medications, or massage)				
Pain		No difference	Moderate (6 RCTs)	0 to 100 scale Short-term: WMD, -2.76 (CI, -5.19 to -0.32; $I^2 = 27%$) 6 mo: WMD, -3.07 (CI, -5.42 to -0.71; $I^2 = 0%$) 12 mo: WMD, -0.76 (CI, -3.19 to 1.66; $I^2 = 0%$)
Function		No difference	Moderate (6 RCTs)	1 mo: SMD, -0.17 (CI, -0.29 to -0.06) 6 mo: SMD, -0.12 (CI, -0.23 to 0.00) 12 mo: SMD, -0.06 (CI, -0.16 to 0.05)
TENS vs. acupuncture				
Pain		No difference	Low (4 RCTs)	Short-term: SMD, 0.15 (CI, -0.33 to 0.63) Long-term: SMD, 0.32 (CI, -0.33 to 0.96)
Lumbar supports vs. other active treatments (traction, spinal manipulation, exercise, physiotherapy, or TENS)				
Pain		No difference	Low (4 RCTs)	No clear differences
Function		No difference	Low (4 RCTs)	

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Appendix Table 2—Continued

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Kinesio taping vs. exercise therapy				
Pain		No difference	Low (2 RCTs)	No clear differences
Function		No difference	Low (2 RCTs)	
Combination therapy treatments vs. monotherapy treatments or no treatment				
MCE plus exercise vs. exercise alone				
Pain		No difference	Low (2 RCTs)	No clear differences
Psychological therapy + exercise or physiotherapy vs. exercise or physiotherapy alone				
Pain		No difference	Low (6 RCTs)	No clear differences
Function		No difference	Low (6 RCTs)	
Spinal manipulation plus another active treatment vs. the other treatment alone				
Pain		Small	Low (3 RCTs)	0 to 100 scale 1 mo: WMD, −5.88 (CI, −10.85 to −0.90) 3 mo: MD, −7.23 (CI, −11.72 to −2.74) 12 mo: MD, −3.31 (CI, −6.60 to −0.02)
Function		Improved	Low (3 RCTs)	0 to 100 scale 1 mo: SMD, −0.40 (CI, −0.73 to −0.07) 3 mo: SMD, −0.22 (CI, −0.38 to −0.06) 12 mo: SMD, −0.21 (CI, −0.34 to −0.09)
Lumbar support plus exercise vs. exercise alone (muscle strengthening)				
Pain		No difference	Low (1 RCT)	No difference in short-term (8 wk) or long-term (6 mo)
Function		No difference	Low (1 RCT)	

CBT = cognitive behavioral therapy; LLLT = low-level laser therapy; MCE = motor control exercise; MD = mean difference; NSAID = nonsteroidal anti-inflammatory drug; NRS = numerical rating scale; ODI = Oswestry Disability Index; OR = odds ratio; RCT = randomized, controlled trial; RDQ = Roland Morris Disability Questionnaire; RR = relative risk; SMD = standardized mean difference; SMR = skeletal muscle relaxant; SSRI = selective serotonin reuptake inhibitor; TCA = tricyclic antidepressant; TENS = transcutaneous electrical nerve stimulation; VAS = visual analogue scale; WMD = weighted mean difference.

* Although some RCTs found that long-acting opioids were associated with greater pain relief, patients randomly assigned to long-acting opioids also received higher doses.

Appendix Table 3. Pharmacologic and Nonpharmacologic Treatments for Radicular Low Back Pain

Intervention	Outcome	Magnitude of Effect	Strength of Evidence (Studies)	Data
Pharmacologic treatments vs. placebo				
Diazepam (acute or subacute radicular pain)				
Pain		Lower likelihood of $\geq 50\%$ improvement	Low (1 RCT)	5 mg twice daily for 5 d: 41% vs. 79%; RR, 0.5 (95% CI, 0.3–0.8)
Function		No effect	Low (1 RCT)	RDQ: No difference through 1 y of follow-up
Systemic corticosteroids				
Pain		No effect	Moderate (6 RCTs)	No clear effect
Function		Small to no effect	Moderate (6 RCTs)	
Nonpharmacologic treatments vs. sham, no treatment, or usual care (acute or subacute)				
Exercise vs. usual care				
Pain		Small	Low (3 RCTs)	Favored exercise, although effects were small
Function		Small	Low (3 RCTs)	
Comparative benefits of pharmacologic and nonpharmacologic treatments				
Traction vs. other treatments				
Pain		No difference	Low (15 RCTs)	No clear differences
Function		No difference	Low (15 RCTs)	
Traction vs. other type of traction				
Pain		No difference	Low (5 RCTs)	No clear differences
Function		No difference	Low (5 RCTs)	
Combination therapy vs. monotherapy or no treatment				
Traction + physiotherapy vs. physiotherapy alone				
Pain		No difference	Low (5 RCTs)	No clear differences
Function		No difference	Low (5 RCTs)	

RCT = randomized, controlled trial; RDQ = Roland Morris Disability Questionnaire; RR = relative risk.

Appendix Table 4. Adverse Events for Treatments for Acute, Chronic, and Radicular Low Back Pain

Interventions	Data on Adverse Events (Quality of Evidence; Studies)	Adverse Effects
Adverse events reported for pharmacologic treatments		
Acetaminophen	Versus placebo: No difference in risk for serious adverse events (moderate quality; 1 RCT) Versus NSAIDs: A systematic review found that acetaminophen was associated with lower risk for adverse events; RR, 0.57 (95% CI, 0.36–0.89) (moderate quality; 3 RCTs)	Thrombocytopenia, agranulocytosis, pancytopenia, hemolytic anemia, methemoglobinemia, hypoglycemia, hypothermia, pancreatitis, nephrotoxicity, hepatotoxicity (with overdose), hepatic necrosis, pneumonitis, rash, and hypersensitivity
NSAIDs	Versus placebo: NSAIDs associated with more adverse effects; RR, 1.35 (CI, 1.09–1.68) (moderate quality; 10 RCTs)	Abdominal pain or cramps, dyspepsia, diarrhea, gastrointestinal bleeding, gastrointestinal perforation, dizziness, headache, edema, rash, heartburn, tinnitus, and pruritus
COX-2-selective NSAIDs	Versus nonselective NSAIDs: COX-2-selective NSAIDs associated with lower risk for adverse effects; RR, 0.83 (CI, 0.70–0.99) (moderate quality; 4 RCTs)	Abdominal pain, diarrhea, dizziness, dyspepsia, edema, flatulence, headache, nausea, rash, upper respiratory tract infection, influenza-like illness, and musculoskeletal and connective tissue signs and symptoms (back pain, muscle spasms, and musculoskeletal pain)
Opioids	Versus placebo: Short-term use associated with higher risk; risks higher in trials that did not use an enriched enrollment and withdrawal design; trials were not designed to assess risks for overdose, abuse, and addiction or long-term harms (moderate quality; 16 studies)	Short-term use: Nausea, dizziness, constipation, vomiting, somnolence, and dry mouth Long-term use: Addiction, abuse, overdose, fractures, cardiovascular events, sexual dysfunction, and motor vehicle accidents
SMRs	Versus placebo (any adverse event): SMRs associated with increased risk; RR, 1.50 (CI, 1.14–1.98) (moderate quality; 8 RCTs) Versus placebo (central nervous system events): SMRs associated with increased risk (primarily sedation); RR, 2.04 (CI, 1.23–3.37) (moderate quality; 8 RCTs)	Sedation, drowsiness, and dizziness
Benzodiazepines	Versus placebo: Central nervous system adverse events reported more frequently with benzodiazepines, although harms were not reported well; no trial was designed to evaluate risks with long-term use (low quality; 9 RCTs)	Somnolence, fatigue, lightheadedness, addiction, abuse, overdose, and fractures
Antidepressants	Versus placebo: Antidepressants associated with higher risk for any adverse events but no differences in rates of specific adverse events or serious adverse events (moderate quality; 12 RCTs) Duloxetine associated with nausea and increased risk for withdrawal due to adverse event	Drowsiness, dizziness, dry mouth, constipation, sexual dysfunction, and nausea
Systemic corticosteroids	Versus placebo: Trials did not report serious adverse events, but adverse events were not reported well in some trials (low quality; 12 RCTs)	Hyperglycemia requiring medical treatment, facial flushing, infection, and gastrointestinal bleeding
Adverse events reported for nonpharmacologic treatments*		
Exercise, Tai chi, massage, and spinal manipulation: Harms typically related to muscle soreness and/or small increases in pain were reported.		
Yoga: Reporting was suboptimal, but almost all adverse events were classified as mild to moderate.		
TENS: Evidence was limited but suggests an increased risk for skin reactions without an increased risk for serious adverse events.		
Heat: Heat was not associated with increased risk for skin flushing vs. no heat or placebo in 2 trials.		

COX-2 = cyclooxygenase-2; NSAID = nonsteroidal anti-inflammatory drug; RCT = randomized, controlled trial; RR = relative risk; SMR = skeletal muscle relaxant; TENS = transcutaneous electrical nerve stimulation.

* Harms were poorly reported in most trials of nonpharmacologic interventions. No serious adverse events were reported.



Clinical practice guidelines for the management of non-specific low back pain in primary care: an updated overview

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Abstract

Objective The aim of this study was to provide an overview of the recommendations regarding the diagnosis and treatment contained in current clinical practice guidelines for patients with non-specific low back pain in primary care. We also aimed to examine how recommendations have changed since our last overview in 2010.

Method The searches for clinical practice guidelines were performed for the period from 2008 to 2017 in electronic databases. Guidelines including information regarding either the diagnosis or treatment of non-specific low back pain, and targeted at a multidisciplinary audience in the primary care setting, were considered eligible. We extracted data regarding recommendations for diagnosis and treatment, and methods for development of guidelines.

Results We identified 15 clinical practice guidelines for the management of low back pain in primary care. For diagnosis of patients with non-specific low back pain, the clinical practice guidelines recommend history taking and physical examination to identify red flags, neurological testing to identify radicular syndrome, use of imaging if serious pathology is suspected (but discourage routine use), and assessment of psychosocial factors. For treatment of patients with acute low back pain, the guidelines recommend reassurance on the favourable prognosis and advice on returning to normal activities, avoiding bed rest, the use of nonsteroidal anti-inflammatory drugs (NSAIDs) and weak opioids for short periods. For treatment of patients with chronic low back pain, the guidelines recommend the use of NSAIDs and antidepressants, exercise therapy, and psychosocial interventions. In addition, referral to a specialist is recommended in case of suspicion of specific pathologies or radiculopathy or if there is no improvement after 4 weeks. While there were a few discrepancies across the current clinical practice guidelines, a substantial proportion of recommendations was consistently endorsed. In the current review, we identified some differences compared to the previous overview regarding the recommendations for assessment of psychosocial factors, the use of some medications (e.g., paracetamol) as well as an increasing amount of information regarding the types of exercise, mode of delivery, acupuncture, herbal medicines, and invasive treatments.

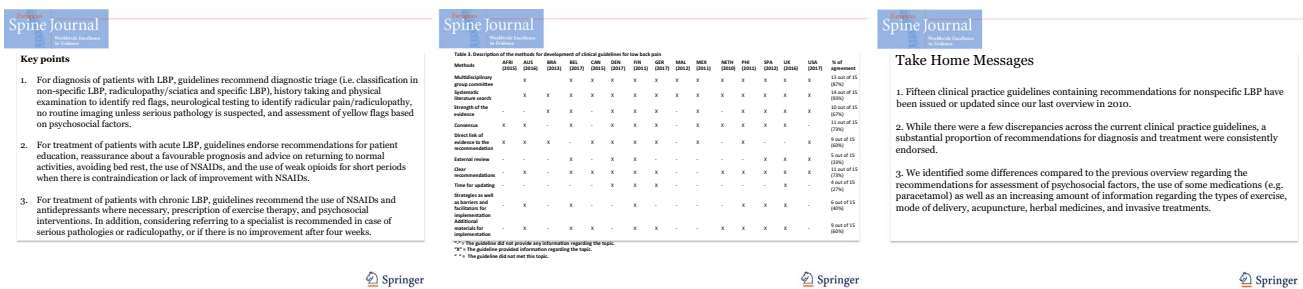
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Extended author information available on the last page of the article

Graphical abstract These slides can be retrieved under Electronic Supplementary Material.



Keywords Low back pain · Clinical guidelines · Diagnosis · Treatment

Introduction

Low back pain (LBP) is the leading contributor to years lived with disability [14]. Non-specific LBP is defined as low back pain not attributable to a known cause [21] and represents 90–95% of the cases of LBP [4]. The estimated point prevalence of non-specific LBP is 18% [13]. Annually, total costs of LBP are estimated to be US \$100 billion in the USA [8], €3.5 billion in the Netherlands [19], €6.6 billion in Switzerland [35], €17.4 billion in Germany [5], and AUD \$9.17 billion in Australia [34]. Although LBP imposes an enormous economic burden on healthcare systems, this condition is responsible to affect individuals' daily lives. Hence, effective strategies play an important role to minimize the impact of LBP.

Clinical practice guidelines provide evidence-based recommendations to assist decision making about health interventions. These documents, developed by expert panels, are normally updated every 3 to 5 years or if the available evidence suggests a reformulation of the previous document is necessary [33]. A brief search of the Central Register of Controlled Trials (CENTRAL) reveals that the number of randomized controlled trials in LBP has nearly doubled since 2010. This finding suggests that some recommendations of clinical practice guidelines for the management of LBP may have changed in recent years.

Since 2001, we have been conducting overviews of clinical practice guidelines for the management of patients with non-specific LBP in primary care settings [17, 18]. These overviews have summarized the overall consensus messages, any differences between clinical practice guidelines, the scientific support for the recommendations, and changes in recommendations over time. The importance of these publications is evidenced by the number of citations received; *Web of Science citation index* notes that the 2001 review [18] was cited 377 times and the 2010 review [17] 316 times. It has been 8 years since our last review and some of the

recommendations for the management of low back pain have likely changed. Therefore, the primary aim of this study was to provide an overview of the recommendations regarding the diagnosis and treatment of patients with non-specific LBP in primary care in current international clinical practice guidelines. We also aimed to examine if recommendations have changed since our last overview.

Methods

Searches

The searches for clinical guidelines were performed for the period from 2008 to 2017 in the following databases: MEDLINE via OVID (key words: combination of search terms regarding low back pain AND clinical guidelines), PEDro (key words: low back pain AND practice guidelines), National Guideline Clearinghouse (www.guideline.gov; key word: low back pain), and National Institute for Health and Clinical Excellence (NICE) (www.nice.org.uk; key word: low back pain). We also checked the guidelines included in our previous review for updates. Furthermore, we conducted citation tracking in the content and reference lists of relevant reviews on guidelines, completed a search of Web of Science citation index for articles citing the previous reviews, and asked experts in the field. Two authors (C.B.O. and C.G.M.) independently screened titles and abstracts of the search results. In case of disagreement, a third author (B.W.K.) arbitrated.

Types of study included

Guidelines including information regarding either the diagnosis or treatment of non-specific LBP, and targeted at a multidisciplinary audience in the primary care setting, were considered eligible. Only guidelines available in English,

French, German, Portuguese, Spanish, Chinese, or Dutch were included because the authors can read these languages. For languages beyond these, we included English language summaries of the guideline if they contained sufficient information. We included one guideline per country unless there were separate guidelines for acute and chronic LBP. We also included guidelines issued by a multinational committee (e.g., Africa, Europe). If more than one guideline was considered eligible, we included the most recent issued by a national body (e.g., national pain society, or national health body).

Data extraction and data synthesis

Two independent authors extracted the following data using a standardised form: recommendations regarding diagnosis and treatment, target population, committee membership, the evidence base of the recommendations (e.g., literature search, grade of evidence), consensus methods (e.g., committee meetings, discussion groups), and dissemination of guidelines (e.g., publication in website or scientific journals). To examine changes in recommendations over time, we compared results of the previous overviews with the current review. We presented the recommendations from the included guidelines in tables.

Results

Electronic searches conducted on June 16, 2017 retrieved 1611 records after removing duplicates. After the screening of titles and abstracts, we assessed 61 full texts against our inclusion criteria. Of these, we excluded 46 full texts because they were: not the most recent guideline issued ($n=19$), not guidelines ($n=15$), not targeted at a multidisciplinary audience ($n=10$), and not in a language where we could obtain a translation ($n=2$). Finally, 15 clinical practice guidelines [1, 3, 7, 9–11, 15, 20, 24, 27, 30–32] for the management of LBP were included from the following countries: Africa (multinational), Australia, Brazil, Belgium, Canada, Denmark, Finland, Germany, Malaysia, Mexico, the Netherlands, Philippines, Spain, the USA, and the UK.

Six guidelines [1, 7, 11, 20, 26, 28] (40%) provided recommendations for patients with acute, subacute, and chronic LBP (i.e., Canada, Finland, Mexico, Philippines, Spain, and the USA), two guidelines [15, 31] (13%) focussed on acute and chronic LBP (i.e., Malaysia and the Netherlands), three guidelines [9, 25, 30] (20%) focussed on acute LBP (i.e., Australia, and Denmark), and one guideline [3] (7%) focussed on chronic LBP (i.e., Brazil). In addition, three guidelines [10, 24, 32] (20%) provided recommendations regardless of the duration of symptoms (i.e., Africa, Belgium, Germany and the UK). Therefore, ten guidelines

contained recommendations for patients with acute LBP, six guidelines contained recommendations for patients with subacute LBP, and nine guidelines contained recommendations for patients with chronic LBP.

Three guidelines [1, 11, 28] defined acute LBP as less than 4 weeks duration, two guidelines [6, 26] specified less than 6 weeks duration and four guidelines [15, 25, 30, 31] defined acute LBP as less than 12 weeks duration. The Canadian guideline [7] defined acute and subacute LBP as less than 12 weeks duration but without specifying the cutoffs for each one. All guidelines defined chronic LBP as more than 12 weeks' duration.

Diagnostic recommendations

Table 1 describes the recommendations regarding diagnosis endorsed by each clinical practice guideline, and “supplementary material: Appendix 1” details these recommendations. Fourteen guidelines provided at least one recommendation regarding diagnosis of patients with LBP. The American guideline [28] did not provide any recommendation regarding diagnosis because the committee group was instructed to make only recommendations for treatment of LBP.

Recommendations for diagnostic triage were found in 13 guidelines. Over half of guidelines [1, 7, 24–26, 31, 32] (7 out of 13; 54%) recommend diagnostic triage to classify patients into one of three categories: non-specific LBP, radiculopathy/sciatica or specific LBP. Almost half of the guidelines [3, 9–11, 15, 20] (46%) recommend the classifications of non-specific LBP and specific LBP without distinguishing the group of patients with radicular pain/radiculopathy. Most guidelines [1, 7, 11, 15, 20, 24–26, 31, 32] (10 out of 12; 83%) recommend history taking and physical examination to identify patients with specific conditions as the cause of the LBP. Box 1 describes the red flags endorsed by most clinical practice guidelines to identify serious conditions in the assessment. In addition, most guidelines [1, 7, 11, 15, 25, 26, 31] (7 out of 9; 78%) recommend neurologic examination to identify radicular pain/radiculopathy including straight leg raise test [1, 7, 15, 26, 32] and assessment of strength, reflexes, and sensation [1, 11, 15]. Only three guidelines [11, 15, 26] (3 out of 12; 25%) recommend an assessment that also includes palpation, posture assessment, and spinal range of movement testing.

All guidelines recommend against the use of routine imaging for patients with non-specific LBP. Most guidelines [1, 7, 9–11, 25, 30] (7 out of 12; 58%) recommend that imaging should only be considered if red flags are present. In addition, five guidelines [1, 7, 10, 24, 32] (42%) suggest imaging when the results are likely to change or direct the treatment (e.g., invasive treatments), and two guidelines

Table 1 Recommendations of clinical guidelines for diagnosis of low back pain

Recommendations for diagnosis	AFRI (2015)	AUS (2016)	BRA (2013)	BEL (2017)	CAN (2015)	DEN (2017)	FIN (2011)	GER (2017)	MAL (2012)	MEX (2011)	NETH (2010)	PHI (2011)	SPA (2012)	UK (2016)	USA (2017)	% of agreement
<i>Diagnostic triage into non-specific LBP; radiculopathy; and specific LBP.</i>		X		X	X	-	X				X	X		X	-	7 out of 13 (54%)
<i>Diagnostic triage into non-specific LBP; and specific LBP.</i>	X		X			-		X	X	X			X		-	6 out of 13 (46%)
<i>History taking and physical examination to identify patients with specific diseases</i>		X	-	X	X	-	X		X	X	X	X	X	X	-	10 out of 12 (83%)
Neurologic examination to identify radicular pain		X	-		X	-	X		X	X	X	X			-	7 out of 9 (78%)
<i>Against the use of routine imaging</i>	X	X	-	X	X	X	X	X	-	X	X	X	X	X	-	12 out of 12 (100%)
Imaging only if serious pathology is suspected	X	X	-		X	X		X	-	X		X			-	7 out of 12 (58%)
Imaging only when the results are likely to change or direct the treatment			-	X	X			X	-			X		X	-	5 out of 12 (42%)
Imaging only if pain persists beyond a period			-		X		X		-						-	2 out of 12 (17%)
<i>Assessment of psychosocial factors based on a list provided by the guideline</i>	X		-		X	-	X		X	X	X	X	X		-	8 out of 12 (67%)
<i>Assessment of psychosocial factors using validated prognostic screening</i>		X	-	X		-		X						X	-	4 out of 12 (33%)
<i>Against the assessment of psychosocial factors using validated prognostic screening</i>					X											1 out of 12 (8%)
<i>Assessment of yellow flags during the first or second consultation</i>		X	-	X	X	-		X	X	X				X	-	7 out of 12 (58%)

“-“ = The guideline did not provide any recommendation regarding the approach.

“X“ = The guideline endorsed the recommendation regarding the approach.

“ “ = The guideline did not endorse the recommendation regarding the approach.

(17%) recommend imaging if pain persists beyond 4 to 6 weeks [7, 26].

Twelve guidelines contain recommendations for assessment of psychosocial factors, or yellow flags, to identify patients with poor prognosis and guide treatment. Most guidelines [1, 7, 9, 11, 15, 20, 26, 31] (8 out of 12; 67%) recommend the assessment based on a list of yellow flags reported in the guideline. Box 2 provides these yellow flags endorsed by most clinical practice guidelines. Four guidelines [10, 24, 25, 32] (33%) recommend assessment using

validated prognostic screening tools (e.g., STarT Back and Orebro) which combine a number of yellow flags. The Danish guideline [30] recommends against targeted treatment for a subgroup of patients with specific prognostic factors. Regarding the optimal timing to assess yellow flags, most guidelines [7, 10, 11, 15, 24, 25, 32] (7 out of 12; 58%) recommend assessment during the first or second consultation.

Box 1 Red flags endorsed by most clinical practice guidelines

Malignancy	History of malignancy (e.g., cancer, neoplasm) [1, 7, 9–11, 15, 20, 24–26, 31, 32], Unexpected weight loss [1, 7, 9–11, 15, 25, 31, 32]
Fracture	Significant trauma [1, 7, 9, 11, 15, 24, 25, 31], prolonged use of corticosteroid [1, 9–11, 15, 20, 25, 31, 32]
Infection	Fever [1, 7, 9–11, 15, 20, 32], HIV [1, 7, 9, 11, 15, 20, 32]

Box 2 Yellow flags endorsed by most clinical practice guidelines

Beliefs that pain and activity are harmful [1, 7, 9, 11, 15, 20, 25, 26, 31, 32]
Treatment preferences that do not fit with the best practice (e.g., passive over active treatments) [1, 7, 9, 15, 20, 25, 26, 31, 32]
Lack of social support [1, 7, 9, 11, 15, 20, 25, 26]

Treatment recommendations

Table 2 provides the recommendations regarding treatment endorsed by each clinical practice guideline, and “supplementary material: Appendix 2” details these recommendations. All guidelines provided at least one recommendation regarding the treatment of LBP.

Recommendations regarding bed rest were provided in 12 guidelines. Most guidelines [7, 9, 11, 15, 25, 30, 31] (7 out of 11; 64%) recommend avoiding bed rest for patients with acute LBP, and four guidelines [1, 10, 20, 26] (36%) recommend for any duration of symptoms. The only exception was the Belgian guideline [32] (8%) which notes an absence of evidence on the benefits or harms of bed rest when used in the short term.

Recommendations on reassurance or advice for patients with non-specific LBP were identified in 14 guidelines. Most guidelines (7 out of 12; 58%) recommend advice to maintain normal activities for patients with acute LBP [1, 7, 10, 15, 25, 30, 32], and some guidelines (42%) recommend the same advice for patients with any duration of symptoms [20, 24, 26, 31, 32]. In addition, most guidelines (10 out of 14; 71%) recommend reassuring the patient that LBP is not a serious illness regardless of the duration of symptoms or reassuring patients with acute LBP of the favorable prognosis [7, 15, 20, 24–26, 28, 30–32].

The recommendations for the prescription of medication vary depending on the class of medication and symptom duration. Most guidelines (14 out of 15; 93%) recommend the use of nonsteroidal anti-inflammatory drugs (NSAIDs) for patients with acute and chronic LBP considering the risk of adverse events (e.g., renal, cardiovascular, and gastrointestinal) [1, 3, 7, 15, 24–26, 28, 32]. For paracetamol/acetaminophen, while most guidelines recommend in favor of this medication [1, 3, 7, 11, 15, 20, 26, 31] (8 out of 14; 57%), five guidelines [10, 24, 27, 30, 32] (36%) advise against the use of paracetamol. The Australian guideline [25] recommends the use of paracetamol but advises that clinicians and patients should be made aware that the medicine might not be effective. Most guidelines (13 out of 15; 87%)

recommend weak opioids [1, 15, 24, 26, 31, 32] for short periods [3, 7, 10, 20, 31, 32], if there is no improvement with NSAIDs or other treatments. The guidelines recommend opioids for acute LBP [1, 7, 9–11, 24, 26, 32] (8 out of 13; 61%), chronic LBP [1, 3, 10, 27, 31] (38%), and for any symptom duration [15, 20] (23%). For antidepressants, most guidelines (6 out of 8; 75%) recommend its use for patients with chronic LBP where necessary [1, 3, 7, 11, 26, 28]. For muscle relaxants, most guidelines [1, 7, 11, 20, 26, 28] (6 out of 11; 54%) recommend this medication for acute LBP [1, 26, 28] (3 out of 6; 50%), chronic LBP [1, 7] (33%), and for any symptom duration [11, 20] (33%). In contrast, five guidelines (5 out of 11; 45%) recommend against muscle relaxants [3, 9, 10, 31, 32]. Two guidelines mentioned the use of herbal medicine for LBP (2 out of 15; 13%); one recommends its use for patients with chronic LBP [7], but the other recommends against it for any type of LBP [10].

Recommendations for referral to a specialist were found in 13 guidelines. Most guidelines [1, 7, 15, 20, 24, 26, 30, 32] (9 out of 13; 69%) recommend referral to a specialist in cases where there is suspicion of serious pathologies or radiculopathy. In addition, most guidelines [7, 9, 10, 20, 25, 30, 31] (7 out of 13; 54%) recommend referral to a specialist if there is no improvement after a time period that ranges from 4 weeks to 2 years.

Recommendations on invasive treatments (e.g., injections, surgery, and radiofrequency denervation) for non-specific LBP were identified in 8 guidelines. Of these, five guidelines (5 out of 8; 62%) recommended against the use of injections for non-specific LBP [7, 10, 24, 25, 31]. In addition, four guidelines [7, 10, 24, 25] (50%) recommend against surgery or radiofrequency denervation [7, 10, 25, 31] (50%) for non-specific LBP. In contrast, three guidelines [1, 24, 32] (37%) recommend radiofrequency denervation for chronic LBP; however, two guidelines [24, 32] (25%) recommended only in strict circumstances such as lack of improvement after conservative treatment, a positive response to a medial branch nerve block, and moderate to severe back pain. Some guidelines recommend surgery for chronic LBP due to disk herniation or spinal instability [1, 15] and common degenerative disorders [1].

Recommendations for multidisciplinary rehabilitation were identified in nine guidelines. Most guidelines (9 out of 11; 90%) recommend multidisciplinary rehabilitation for patients with chronic LBP [7, 10, 11, 15, 24–26, 28, 32]. One guideline [20] recommends multidisciplinary rehabilitation

Table 2 Recommendations of clinical practice guidelines for treatment of low back pain

Recommendations for treatment	AFRI (2015)	AUS (2016)	BRA (2013)	BEL (2017)	CAN (2015)	DEN (2017)	FIN (2011)	GER (2017)	MAL (2012)	MEX (2011)	NETH (2010)	PHI (2011)	SPA (2012)	UK (2016)	USA (2017)	% of agreement
<i>Avoiding bed rest</i>	X	X	-		X	X	X	X	X	X	X	X	X	-	-	11 out of 12 (92%)
Acute LBP	X	X	-		X	X			X	X	X			-	-	7 out of 11 (64%)
Any duration of symptoms			-				X	X				X	X			4 out of 11 (36%)
<i>Using patient education - advise to maintain normal activities</i>		X	-	X	X	X	X	X	X		X	X	X	X	X	12 out of 14 (68%)
Acute LBP		X	-		X	X		X	X			X			X	7 out of 12 (58%)
Any symptom duration			-	X			X				X		X	X	-	5 out of 12 (42%)
<i>Using patient education - reassurance</i>		X	-	X	X	X	X		X		X		X	X	X	10 out of 14 (71%)
<i>Prescription of NSAIDs for any symptom duration</i>	X	X	X	X	X	X	X	X		X	X	X	X	X	X	14 out of 15 (93%)
<i>Insufficient data regarding NSAIDs for chronic LBP</i>									X							1 out of 15 (7%)
<i>Prescription of paracetamol</i>	-		X		X		X		X	X	X	X	X			8 out of 14 (57%)
Acute LBP	-				X		X				X	X				4 out of 8 (50%)
Chronic LBP	-		X		X							X				3 out of 8 (37%)
Any symptom duration	-								X	X			X			3 out of 8 (37%)
<i>Against the prescription of paracetamol</i>	-			X		X		X						X	X	5 out of 14 (36%)
<i>Using opioids</i>	X		X	X	X		X	X	X	X	X	X	X	X	X	13 out of 15 (87%)
Acute LBP	X			X	X		X	X		X		X		X		8 out of 13 (61%)
Chronic LBP			X				X				X	X			X	5 out of 13 (38%)
Any duration of symptoms									X				X			2 out of 13 (23%)
<i>Against the prescription of opioids</i>		X				X								X		3 out of 15 (23%)
Acute LBP		X				X										2 out of 3 (66%)
Chronic LBP														X		1 out of 3 (33%)

for any duration of symptoms, and one guideline [31] recommends if there is no improvement after monodisciplinary approach.

Recommendations for psychosocial strategies were found across eleven guidelines. Most guidelines (10 out of 11; 91%) endorse the use of a cognitive behavior approach [7, 10, 11, 20, 24–26, 28, 31, 32]. In addition, most guidelines (9 out of 11; 82%) recommend these therapies for patients

with chronic LBP [7, 10, 15, 20, 24, 26, 28, 31, 32] with some of them recommending only if psychosocial factors are identified [15, 24, 31, 32].

All clinical practice guidelines provided recommendations for exercise therapy. Most guidelines (10 out of 14; 71%) recommend exercise therapy for patients with chronic LBP [1, 3, 7, 11, 15, 20, 26, 28, 31]. Noteworthy, we identified great discrepancy in the type of exercise program (e.g.,

Table 2 (continued)

<i>Using antidepressants</i>	-	-	X		X	-	X	X	-	X	-	X	X		X	8 out of 10 (80%)
Chronic LBP	-	-	X		X	-	X		-	X	-	X			X	6 out of 8 (75%)
<i>Against the prescription of antidepressants</i>	-	-		X		-			-		-			X		2 out of 10 (20%)
<i>Using muscle relaxants</i>		-			X	-	X		-	X		X	X	-	X	6 out of 11 (54%)
Acute LBP		-				-	X		-			X		-	X	3 out of 6 (50%)
Chronic LBP		-			X	-			-			X		-		2 out of 6 (33%)
Any duration of symptoms		-				-			-	X			X	-		2 out of 6 (33%)
<i>Against the prescription of muscle relaxants</i>	X	-	X	X		-		X	-		X			-		5 out of 11 (45%)
<i>Using herbal medicines</i>	-	-	-	-	X	-	-	-	-	-	-	-	-	-	-	1 out of 2 (50%)
<i>Against the prescription of herbal medicines</i>	-	-	-	-	-	-	-	X	-	-	-	-	-	-	-	1 out of 2 (50%)
<i>Referral to specialist in case of suspicion of specific pathologies or radiculopathy</i>			-	X	X	X	X	X	X			X	X	X	-	9 out of 13 (69%)
<i>Referral to specialist if there is no improvement after four weeks to two years</i>	X	X	-		X	X		X			X		X		-	7 out of 13 (54%)
<i>Against injections</i>	-	X	-		X	-	-	X		-	X		-	X	-	5 out of 8 (62%)
<i>Using surgery</i>	-		-			-	-		X	-		X	-		-	2 out of 8 (25%)
<i>Against surgery</i>	-	X	-		X	-	-	X		-			-	X	-	4 out of 8 (50%)
<i>Using radiofrequency denervation for chronic LBP.</i>	-		-	X		-	-		-			X	-	X	-	3 out of 8 (37%)
<i>Against radiofrequency denervation for nonspecific LBP.</i>	-	X	-		X	-	-	X		-	X		-		-	4 out of 8 (50%)
<i>Using multidisciplinary rehabilitation</i>	-	X	-	X	X	-	X	X	X	X	X	-	X	X	X	11 out of 11 (100%)
Chronic LBP	-	X	-	X	X	-	X	X	X	X		-		X	X	9 out of 11 (81%)
Any duration of symptoms	-		-			-						-	X			1 out of 11 (9%)
Patients not recovered after monodisciplinary approach	-		-			-					X	-				1 out of 11 (9%)

aquatic exercises, stretching, back schools, McKenzie exercise approach, yoga, and tai-chi) and mode of delivery (e.g., individually designed programs, supervised home exercise, and group exercise). Guidelines provided inconsistent recommendations on exercise therapy for acute LBP.

The recommendations for spinal manipulation and acupuncture vary across clinical practice guidelines. Eleven guidelines provided recommendations for spinal manipulation, and nine guidelines recommended its use. Most guidelines (6 out of 9; 66%) recommend spinal manipulation for

Table 2 (continued)

<i>Using psychosocial therapy</i>	-	X	-	X	X	-	X	X	X	X	X	-	X	X	X	11 out of 11 (100%)
Chronic LBP	-		-	X	X	-	X	X	X		X	-	X	X	X	9 out of 11 (82%)
Acute LBP	-	X	-			-						-				1 out of 11 (9%)
Any duration of symptoms	-		-			-				X		-				1 out of 11 (9%)
<i>Using exercise therapy</i>	X		X	X	X	X	X	X	X	X	X	X	X	X	X	14 out of 15 (93%)
Chronic LBP			X		X		X	X	X	X	X	X	X		X	10 out of 14 (71%)
Acute LBP	X					X					X					3 out of 14 (21%)
Any duration of symptoms				X										X		2 out of 14 (14%)
<i>Using spinal manipulation</i>	-	-	X	X	X		X	X	-	X	X	-	X	X		9 out of 11 (81%)
Acute LBP	-	-		X	X		X		-	X	X	-			X	6 out of 9 (66%)
Chronic LBP							X					X			X	3 out of 9 (33%)
Any duration of symptoms				X				X						X		3 out of 9 (33%)
<i>Against the use of spinal manipulation</i>	X	-	-						-	X		-				2 out of 11 (19%)
Chronic LBP										X						1 out of 2 (50%)
Acute LBP	X															1 out of 2 (50%)
<i>Using acupuncture</i>	X	X	-	-		X	-		-	-	-	-			X	4 out 8 (50%)
<i>Against the use of acupuncture</i>			-	-	X		-	X	-	-	-	X	-	X		4 out 8 (50%)

“-“ = The guideline did not provide any recommendation regarding the approach.

“X“ = The guideline endorsed the recommendation regarding the approach.

“ “ = The guideline did not endorse the recommendation regarding the approach.

acute LBP, but there are some discrepancies on the indications. The guidelines recommend spinal manipulation in addition to usual care [30], if there is no improvement after other treatments [7, 31], or in any circumstance [10, 28]. Three guidelines [15, 24, 32] (33%) recommend spinal manipulation as a component of a multimodal or active treatment program for patients with any symptom duration. Three guidelines (33%) recommend spinal manipulation as a component of a multimodal treatment program [10] or in any circumstance for chronic LBP [28]. In contrast, two guidelines recommend against spinal manipulation for acute LBP [9] or chronic LBP [31].

Similarly, the recommendations for acupuncture were inconsistent. Four guidelines [1, 7, 10, 28] recommend the use of acupuncture. Of these, three guidelines recommend acupuncture for patients with acute and chronic LBP [1, 28]. One guideline [7, 10] recommends acupuncture as an adjunct of an active rehabilitation program for patients with

chronic LBP. Four out of eight guidelines do not recommend acupuncture [9, 24, 30] (37%) or state that acupuncture should be avoided [25] (13%).

Methods of development of the clinical practice guidelines

Table 3 provides the methods of development and implementation reported by each clinical practice guideline, and “supplementary material: Appendix 3” details these methods. Most guidelines [1, 7, 10, 11, 15, 20, 24–26, 28, 30–32] were issued by a multidisciplinary group including healthcare professionals such as primary care physicians, physical and manual therapists, chiropractors, psychologists, orthopaedic surgeons, rheumatologists, and radiologists. The African guideline [9] was developed by a medical group, and the Brazilian guideline [3] was developed by an association comprised of physiatrists.

Table 3 Description of the methods for development of clinical guidelines for low back pain

Methods	AFRI (2015)	AUS (2016)	BRA (2013)	BEL (2017)	CAN (2015)	DEN (2017)	FIN (2011)	GER (2017)	MAL (2012)	MEX (2011)	NETH (2010)	PHI (2011)	SPA (2012)	UK (2016)	USA (2017)	% of agreement
<i>Multidisciplinary group committee</i>	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	13 out of 15 (87%)
<i>Systematic literature search</i>	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	14 out of 15 (93%)
<i>Strength of the evidence</i>	–	–	X	X	–	X	X	X	–	X	–	X	X	X	X	10 out of 15 (67%)
<i>Consensus</i>	X	X	–	X	–	X	X	X	–	X	X	X	X	X	–	11 out of 15 (73%)
<i>Direct link of evidence to the recommendation</i>	X	X	X	–	X	X	X	X	–	X	–	X	–	–	X	9 out of 15 (60%)
<i>External review</i>	–	–	–	X	–	X	X	–	–	–	–	–	X	X	X	5 out of 15 (33%)
<i>Clear recommendations</i>	–	X	–	X	X	X	X	X	–	–	X	X	X	X	X	11 out of 15 (73%)
<i>Time for updating</i>	–	–	–	–	–	X	X	X	–	–	–	–	–	X	–	4 out of 15 (27%)
<i>Strategies as well as barriers and facilitators for implementation</i>	–	X	–	X	–	–	X	–	–	–	–	X	X	X	–	6 out of 15 (40%)
<i>Additional materials for implementation</i>	–	X	–	X	X	–	X	X	–	–	X	X	X	X	–	9 out of 15 (60%)

“–” The guideline did not provide any information regarding the topic

“X” The guideline provided information regarding the topic

“.” The guideline did not met this topic

Most guidelines based their recommendations on systematic literature searches of electronic databases and previous version of guidelines (14 out of 15; 93%) [1, 3, 7, 10, 11, 15, 20, 24–26, 28, 30–32], evaluated the strength of the evidence (10 out of 15; 67%) [1, 3, 10, 11, 20, 24–28, 30, 32], and used consensus in the working group when necessary (11 out of 15; 73%) [1, 9–11, 20, 24–26, 30–32]. In addition, most guidelines gave direct links between the recommendations and the evidence (9 out of 15; 60%) [1, 3, 7, 9–11, 25, 30] and provided clear and specific recommendations (11 out of 15; 73%) [1, 7, 10, 20, 24–26, 28, 30–32]. In contrast, few guidelines provided sufficient information regarding their external review process (5 out of 15; 33%) [20, 24, 28, 30, 32] and the time frame for updates (4 out of 15; 27%) [10, 24, 26, 30]. Where it was reported, this ranged from 2 to 5 years.

Most guidelines were available on the website of the participating organization, and some guidelines [3, 10, 11, 28, 30] were published in scientific journals. Most guidelines (9 out of 15; 60%) were accompanied by additional materials for dissemination [1, 7, 10, 20, 24–26, 31, 32] such as different versions for patients and clinicians, a care pathway, a summary version, an interactive flowchart, or videos. A few guidelines (6 out of 15; 40%) reported strategies or the barriers and facilitators for implementation [1, 20, 24, 26, 32].

Changes in recommendations over time

Few changes were identified in the recommendations on diagnosis of non-specific LBP compared to the previous guidelines. Currently, most guidelines still recommend the assessment of psychosocial factors based on yellow flags at the first or second consultation. Of note, an increasing proportion (33%) of guidelines are recommending the use of validated prognostic screening tools (e.g., STarT Back screening tool or Örebro).

Some recommendations changed compared to the previous guidelines for the use of medications for non-specific LBP. Our 2010 overview found a hierarchical order including paracetamol as the first choice and NSAIDs as the second choice. In this review, we identified that most guidelines recommend only the use of NSAIDs as the first choice for any duration of symptoms. Of note, most current guidelines recommend antidepressants, where necessary, for chronic LBP which was not endorsed by the previous guidelines. The recommendations regarding the NSAIDs and antidepressants were consistent across guidelines included in this review.

We also identified more details on the recommendations regarding some approaches compared to the past guidelines. The current clinical practice guidelines suggest some types of exercise and modes of delivery for patients with chronic LBP compared to the previous guidelines which only noted the preference for using intensive training. We also found

recommendations regarding some approaches in this review which were not previously cited in past guidelines such as the use of herbal medicines, acupuncture, and invasive treatments. However, the recommendations regarding these approaches were inconsistent or cited in a small proportion of guidelines (i.e., less than 50% of the guidelines).

Discussion

Fifteen clinical practice guidelines containing recommendations for non-specific LBP have been issued or updated since our last overview in 2010. For the diagnostic recommendations, guidelines recommend diagnostic triage (i.e., classification in non-specific LBP, radiculopathy/sciatica, and specific LBP), history taking and physical examination to identify red flags, neurological testing to identify radicular pain/radiculopathy, no routine imaging unless serious pathology is suspected, and assessment of yellow flags based on psychosocial factors cited in the guidelines in the first or second evaluation. For treatment of patients with acute LBP, most guidelines endorse recommendations for patient education, reassurance about a favourable prognosis and advice on returning to normal activities, avoiding bed rest, the use of NSAIDs and weak opioids for short periods when there is contraindication or lack of improvement with NSAIDs. For treatment of patients with chronic LBP, most guidelines recommend the use of NSAIDs and antidepressants where necessary, prescription of exercise therapy, and psychosocial interventions. In addition, considering referring to a specialist is recommended in case of serious pathologies or radiculopathy, or if there is no improvement after 4 weeks to 2 years.

Discrepancies in the recommendations across the guidelines

We identified discrepancies in the recommendations for the use of paracetamol, muscle relaxants, and herbal medicines. For paracetamol, the most recent guidelines [10, 24, 28, 30, 32] do not recommend this medication. This change might be attributable to recent studies demonstrating the lack of efficacy of paracetamol for non-specific LBP [29, 36]. In addition, the inconsistent recommendations for the use of muscle relaxants, and herbal medicines might be attributable to different care settings and cultural context across the countries.

Most guidelines recommend the use of weak opioids for short periods if NSAIDs are contraindicated or not effective for patients with acute LBP, despite an absence of relevant clinical trials as demonstrated by a recent systematic review [2]. Considering the rising prescription of opioids [22], the use of this pain medication has been discouraged due to

the small benefit on pain intensity in chronic LBP as well as potential side effects (e.g., misuse or physical dependence) [2, 23]. Although the current review found that most guidelines recommend opioids for acute LBP, this recommendation is not supported by the evidence and may result in increased harms for patients with non-specific LBP.

The recommendations on spinal manipulation and acupuncture are inconsistent but in different aspects. The recommendations on spinal manipulation vary mainly regarding the circumstances in which the intervention should be administered (e.g., any circumstance, in addition to usual care, after lack of improvement). The recommendations on acupuncture have discrepancies related to its use in patients with non-specific LBP. In addition, four guidelines [1, 7, 10, 28] recommend acupuncture, but disagree regarding duration of symptoms. These discrepancies might be attributable to the lack of high-quality evidence which may result in recommendations based on group consensus considering different aspects. Future studies should be conducted to clarify these recommendations.

Few changes in the recommendations over time

Although the number of randomised controlled trials has nearly doubled since 2010, the recommendations regarding management remain similar compared to the previous review. We identified an increasing proportion (33%) of guidelines recommending the assessment of yellow flags using prognostic screening tools [10, 24, 25, 32]. This might be attributable to a recent randomised clinical trial that showed small improvements from targeting treatment based on responses to a validated prognostic screening tool [12]. However, this was based on one study only, and a recent review [16] found that screening tools poorly identify patients who will develop chronic pain and worse outcomes in patients with LBP. Future studies should be conducted before any definitive conclusion can be made regarding the use of prognostic models.

The guidelines still uniformly recommend exercise for chronic LBP. However, the clinical practice guidelines are now suggesting a greater variety of types of exercise. For example, guidelines include options such as sports rehabilitation, physical activity as tolerated, aquatic exercises, stretching, aerobic, strength training, endurance, motor control exercise, yoga, and tai-chi. Although the guidelines endorsing some types of exercise increased [1, 7, 20, 24, 26, 28], there is no consistency in the recommendations favouring one particular modality. Hence, we would argue that the choice may rely on patients' preferences and therapists' experience.

Future developments in research and guideline development

Our overview included clinical practice guidelines that issued recommendations for patients with nonspecific LBP. Although some guidelines also include recommendations for different types of LBP, future studies should investigate the recommendations for radicular pain/radiculopathy and specific LBP. Another limitation of this review is the absence of quality assessment of the guidelines using a validated tool (e.g., AGREE). Nevertheless, we provided an overview of the methods of the clinical practice guidelines included in the current review.

Based on the recommendations for the development of guidelines for LBP provided by the previous review, the methods for developing the guidelines seem to have improved over the years (Box 3). Most guidelines provided a description for obtaining the evidence to be used in the recommendations, with some describing the method for assessing the strength of the evidence (Recommendation 1). However, only two guidelines [20, 30] (13%) included non-English publications (Recommendation 2). The target group and the committee of the guideline were well described (Recommendations 3 and 4). A substantial proportion (53%) of guidelines provided a direct link between the evidence and recommendations (Recommendation 5). Although an increasing number of guidelines reported details regarding the consensus methods, this topic was still not appropriately described by the guidelines (Recommendation 6). One issue that remained over the years was that the clinical practice guidelines did not often incorporate information regarding effectiveness and health benefits as well as the cost-effectiveness (Recommendation 7). As mentioned earlier, the strategies for dissemination of the guidelines have improved substantially with several types of materials available for patients and clinicians. However, although the details regarding implementation also improved, most guidelines did not specify the strategies as well as the barriers and facilitators for implementation in the clinical practice (Recommendation 8). In addition, few guidelines [10, 24, 26, 30] provided the methods and time frame for updating (Recommendation 9).

Conclusion

The current clinical practice guidelines recommend diagnostic triage using history taking and physical examination to identify red flags and neurological testing to identify radicular pain/radiculopathy, against routine imaging unless serious pathology is suspected, and assessment of yellow flags based on psychosocial factors cited in the guidelines in the first or second evaluation. For acute LBP, most guidelines

Box 3 Recommendations for the development of future guidelines in the field of low back pain

1. Make use of available evidence-based reviews and previous clinical guidelines
2. Include relevant non-English publications (if available)
3. Determine in advance the intended target groups (healthcare professions, patient population, and policy makers)
4. Be aware that the makeup of the guideline committee may have a direct impact on the content of the recommendations
5. Specify exactly which recommendations are evidence based and supply the correct references to each of these recommendations
6. Specify exactly which recommendations are consensus based and explain the process
7. Specify effectiveness and cost-effectiveness of the recommendations
8. Determine barriers, facilitators, and action for implementing in clinical practice strategy
9. Provide a time frame for future updates of the guideline

endorsed recommendations for patient education, reassurance about the favourable prognosis and advice on returning to normal activities, avoiding bed rest, the use of NSAIDs and weak opioids for short periods where necessary. For chronic LBP, most guidelines recommended the use of NSAIDs and antidepressants where necessary, prescription of exercise therapy, and psychosocial interventions. In addition, referring to a specialist is recommended in cases where there is suspicion of serious pathologies or radiculopathy or if there is no improvement after 4 weeks to 2 years.

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Compliance with ethical standards

Conflict of interest C.G.M. reports receiving lecture fees from Pfizer. No other conflict of interest relevant to this article was declared.

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